

? A typical runner gains speed gradually during the course of a sprinting foot race and then slows down after crossing the finish line. In which part of the motion is it accurate to say that the runner is accelerating? (i) During the race; (ii) after the runner crosses the finish line; (iii) both (i) and (ii); (iv) neither (i) nor (ii); (v) answer depends on how rapidly the runner gains speed during the race.



2 Motion Along a Straight Line

LEARNING OUTCOMES

In this chapter, you'll learn...

- 2.1 How the ideas of displacement and average velocity help us describe straight-line motion.
- 2.2 The meaning of instantaneous velocity; the difference between velocity and speed.
- 2.3 How to use average acceleration and instantaneous acceleration to describe changes in velocity.
- 2.4 How to use equations and graphs to solve problems that involve straight-line motion with constant acceleration.
- 2.5 How to solve problems in which an object is falling freely under the influence of gravity alone.
- 2.6 How to analyze straight-line motion when the acceleration is not constant.

You'll need to review...

- 1.7 The displacement vector.
- 1.8 Components of a vector.

What distance must an airliner travel down a runway before it reaches takeoff speed? When you throw a baseball straight up in the air, how high does it go? When a glass slips from your hand, how much time do you have to catch it before it hits the floor? These are the kinds of questions you'll learn to answer in this chapter. *Mechanics* is the study of the relationships among force, matter, and motion. In this chapter and the next we'll study *kinematics*, the part of mechanics that enables us to describe motion. Later we'll study *dynamics*, which helps us understand why objects move in different ways.

In this chapter we'll concentrate on the simplest kind of motion: an object moving along a straight line. To describe this motion, we introduce the physical quantities *velocity* and *acceleration*. In physics these quantities have definitions that are more precise and slightly different from the ones used in everyday language. Both velocity and acceleration are *vectors*: As you learned in Chapter 1, this means that they have both magnitude and direction. Our concern in this chapter is with motion along a straight line only, so we won't need the full mathematics of vectors just yet. But using vectors will be essential in Chapter 3 when we consider motion in two or three dimensions.

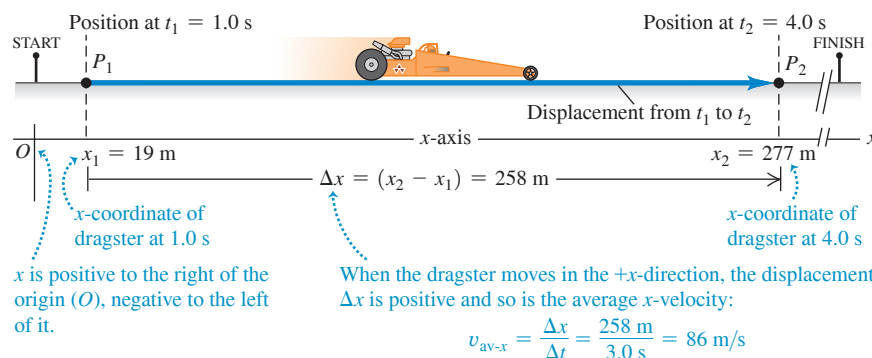
We'll develop simple equations to describe straight-line motion in the important special case when acceleration is constant. An example is the motion of a freely falling object. We'll also consider situations in which acceleration varies during the motion; in this case, it's necessary to use integration to describe the motion. (If you haven't studied integration yet, Section 2.6 is optional.)

2.1 DISPLACEMENT, TIME, AND AVERAGE VELOCITY

Suppose a drag racer drives her dragster along a straight track (**Fig. 2.1**). To study the dragster's motion, we need a coordinate system. We choose the x -axis to lie along the dragster's straight-line path, with the origin O at the starting line. We also choose a point on the dragster, such as its front end, and represent the entire dragster by that point. Hence we treat the dragster as a **particle**.

A useful way to describe the motion of this particle is in terms of the change in its coordinate x over a time interval. Suppose that 1.0 s after the start the front of the dragster is at point P_1 , 19 m from the origin, and 4.0 s after the start it is at point P_2 , 277 m

Figure 2.1 Positions of a dragster at two times during its run.



from the origin. The *displacement* of the particle is a vector that points from P_1 to P_2 (see Section 1.7). Figure 2.1 shows that this vector points along the x -axis. The x -component (see Section 1.8) of the displacement is the change in the value of x , $(277 \text{ m} - 19 \text{ m}) = 258 \text{ m}$, that took place during the time interval of $(4.0 \text{ s} - 1.0 \text{ s}) = 3.0 \text{ s}$. We define the dragster's **average velocity** during this time interval as a *vector* whose x -component is the change in x divided by the time interval: $(258 \text{ m})/(3.0 \text{ s}) = 86 \text{ m/s}$.

In general, the average velocity depends on the particular time interval chosen. For a 3.0 s time interval *before* the start of the race, the dragster is at rest at the starting line and has zero displacement, so its average velocity for this time interval is zero.

Let's generalize the concept of average velocity. At time t_1 the dragster is at point P_1 , with coordinate x_1 , and at time t_2 it is at point P_2 , with coordinate x_2 . The displacement of the dragster during the time interval from t_1 to t_2 is the vector from P_1 to P_2 . The x -component of the displacement, denoted Δx , is the change in the coordinate x :

$$\Delta x = x_2 - x_1 \quad (2.1)$$

The dragster moves along the x -axis only, so the y - and z -components of the displacement are equal to zero.

The x -component of average velocity, or the **average x -velocity**, is the x -component of displacement, Δx , divided by the time interval Δt during which the displacement occurs. We use the symbol $v_{\text{av-}x}$ for average x -velocity (the subscript "av" signifies average value, and the subscript x indicates that this is the x -component):

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \quad (2.2)$$

Average x -velocity of a particle in straight-line motion during time interval from t_1 to t_2

x -component of the particle's displacement

Final x -coordinate minus initial x -coordinate

Time interval

Final time minus initial time

As an example, for the dragster in Fig. 2.1, $x_1 = 19 \text{ m}$, $x_2 = 277 \text{ m}$, $t_1 = 1.0 \text{ s}$, and $t_2 = 4.0 \text{ s}$. So Eq. (2.2) gives

$$v_{\text{av-}x} = \frac{277 \text{ m} - 19 \text{ m}}{4.0 \text{ s} - 1.0 \text{ s}} = \frac{258 \text{ m}}{3.0 \text{ s}} = 86 \text{ m/s}$$

The average x -velocity of the dragster is positive. This means that during the time interval, the coordinate x increased and the dragster moved in the positive x -direction (to the right in Fig. 2.1).

If a particle moves in the *negative* x -direction during a time interval, its average velocity for that time interval is negative. For example, suppose an official's truck moves to the left along the track (Fig. 2.2, next page). The truck is at $x_1 = 277 \text{ m}$ at $t_1 = 16.0 \text{ s}$ and is at $x_2 = 19 \text{ m}$ at $t_2 = 25.0 \text{ s}$. Then $\Delta x = (19 \text{ m} - 277 \text{ m}) = -258 \text{ m}$ and $\Delta t = (25.0 \text{ s} - 16.0 \text{ s}) = 9.0 \text{ s}$. The x -component of average velocity is $v_{\text{av-}x} = \Delta x / \Delta t = (-258 \text{ m}) / (9.0 \text{ s}) = -29 \text{ m/s}$. Table 2.1 lists some simple rules for deciding whether the x -velocity is positive or negative.

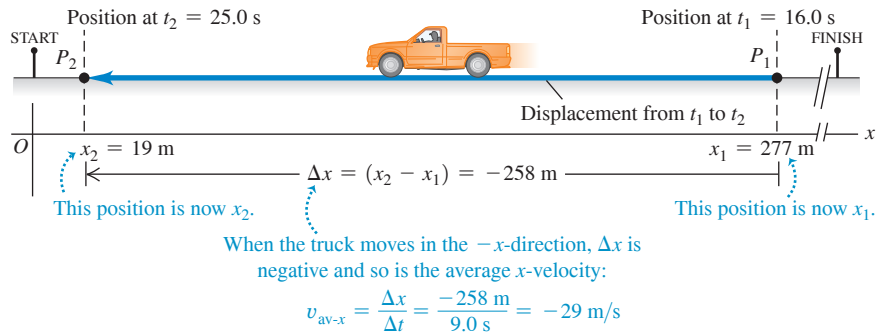
CAUTION The meaning of Δx Note that Δx is *not* the product of Δ and x ; it is a single symbol that means "the change in quantity x ." We use the Greek capital letter Δ (delta) to represent a *change* in a quantity, equal to the *final* value of the quantity minus the *initial* value—never the reverse. Likewise, the time interval from t_1 to t_2 is Δt , the change in t : $\Delta t = t_2 - t_1$ (final time minus initial time). ■

TABLE 2.1 Rules for the Sign of x -Velocity

If x -coordinate is:	... x -velocity is:
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$ -direction
Positive & decreasing (getting less positive)	Negative: Particle is moving in $-x$ -direction
Negative & increasing (getting less negative)	Positive: Particle is moving in $+x$ -direction
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$ -direction

Note: These rules apply to both the average x -velocity $v_{\text{av-}x}$ and the instantaneous x -velocity v_x (to be discussed in Section 2.2).

Figure 2.2 Positions of an official’s truck at two times during its motion. The points P_1 and P_2 now indicate the positions of the truck, not the dragster, and so are the reverse of Fig. 2.1.



CAUTION The sign of average x -velocity In our example positive v_{av-x} means motion to the right, as in Fig. 2.1, and negative v_{av-x} means motion to the left, as in Fig. 2.2. But that’s *only* because we chose the $+x$ -direction to be to the right. Had we chosen the $+x$ -direction to be to the left, the average x -velocity v_{av-x} would have been negative for the dragster moving to the right and positive for the truck moving to the left. In many problems the direction of the coordinate axis is yours to choose. Once you’ve made your choice, you *must* take it into account when interpreting the signs of v_{av-x} and other quantities that describe motion!

With straight-line motion we sometimes call Δx simply the displacement and v_{av-x} simply the average velocity. But remember that these are the x -components of vector quantities that, in this special case, have *only* x -components. In Chapter 3, displacement, velocity, and acceleration vectors will have two or three nonzero components.

Figure 2.3 is a graph of the dragster’s position as a function of time—that is, an **x - t graph**. The curve in the figure *does not* represent the dragster’s path; as Fig. 2.1 shows, the path is a straight line. Rather, the graph represents how the dragster’s position changes with time. The points p_1 and p_2 on the graph correspond to the points P_1 and P_2 along the dragster’s path. Line p_1p_2 is the hypotenuse of a right triangle with vertical side $\Delta x = x_2 - x_1$ and horizontal side $\Delta t = t_2 - t_1$. The average x -velocity $v_{av-x} = \Delta x / \Delta t$ of the dragster equals the *slope* of the line p_1p_2 —that is, the ratio of the triangle’s vertical side Δx to its horizontal side Δt . (The slope has units of meters divided by seconds, or m/s, the correct units for average x -velocity.)

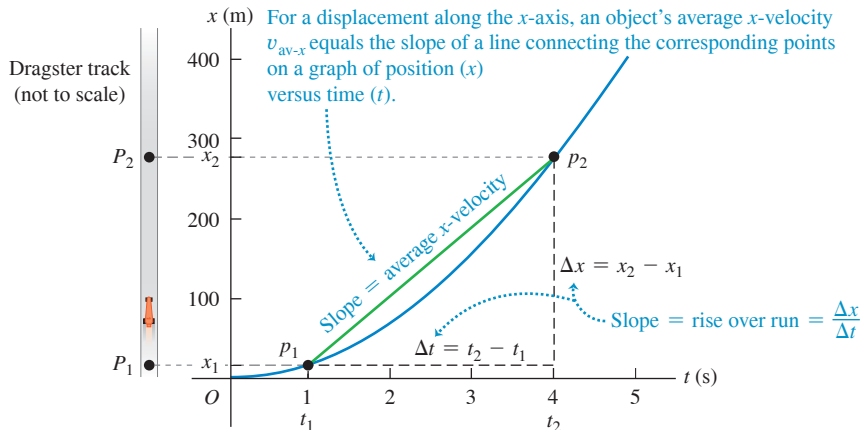
The average x -velocity depends on only the total displacement $\Delta x = x_2 - x_1$ that occurs during the time interval $\Delta t = t_2 - t_1$, not on what happens during the time interval. At time t_1 a motorcycle might have raced past the dragster at point P_1 in Fig. 2.1, then slowed down to pass through point P_2 at the same time t_2 as the dragster. Both vehicles have the same displacement during the same time interval and so have the same average x -velocity.

If distance is given in meters and time in seconds, average velocity is measured in meters per second, or m/s (Table 2.2). Other common units of velocity are kilometers per hour (km/h), feet per second (ft/s), miles per hour (mi/h), and knots (1 knot = 1 nautical mile/h = 6080 ft/h).

TABLE 2.2 Typical Velocity Magnitudes

A snail’s pace	10^{-3} m/s
A brisk walk	2 m/s
Fastest human	11 m/s
Freeway speeds	30 m/s
Fastest car	341 m/s
Random motion of air molecules	500 m/s
Fastest airplane	1000 m/s
Orbiting communications satellite	3000 m/s
Average speed of an electron in a hydrogen atom	2×10^6 m/s
Light traveling in vacuum	3×10^8 m/s

Figure 2.3 A graph of the position of a dragster as a function of time.



TEST YOUR UNDERSTANDING OF SECTION 2.1 Five cars, *A*, *B*, *C*, *D*, and *E*, each take a trip that lasts one hour. The positive *x*-direction is to the east. (i) *A* travels 50 km due east. (ii) *B* travels 50 km due west. (iii) *C* travels 60 km due east, then turns around and travels 10 km due west. (iv) *D* travels 70 km due east. (v) *E* travels 20 km due west, then turns around and travels 20 km due east. (a) Rank the five trips in order of average *x*-velocity from most positive to most negative. (b) Which trips, if any, have the same average *x*-velocity? (c) For which trip, if any, is the average *x*-velocity equal to zero?

ANSWERS

(a) (i) $+50 \text{ km/h}$; (ii) -50 km/h ; (iii) $+50 \text{ km/h}$; (iv) $+70 \text{ km/h}$; (v) -20 km/h .
 (b) Trips (i) and (iii) have the same average *x*-velocity of $+50 \text{ km/h}$.
 (c) Trip (v) has an average *x*-velocity equal to zero.

2.2 INSTANTANEOUS VELOCITY

Sometimes average velocity is all you need to know about a particle's motion. For example, a race along a straight line is really a competition to see whose average velocity, $v_{\text{av-}x}$, has the greatest magnitude. The prize goes to the competitor who can travel the displacement Δx from the start to the finish line in the shortest time interval, Δt (Fig. 2.4).

But the average velocity of a particle during a time interval can't tell us how fast, or in what direction, the particle was moving at any given time during the interval. For that we need to know the **instantaneous velocity**, or the velocity at a specific instant of time or specific point along the path.

CAUTION How long is an instant? You might use the phrase "It lasted just an instant" to refer to something that spanned a very short time interval. But in physics an instant has no duration at all; it refers to a single value of time. ■

To find the instantaneous velocity of the dragster in Fig. 2.1 at point P_1 , we move point P_2 closer and closer to point P_1 and compute the average velocity $v_{\text{av-}x} = \Delta x / \Delta t$ over the ever-shorter displacement and time interval. Both Δx and Δt become very small, but their ratio does not necessarily become small. In the language of calculus, the limit of $\Delta x / \Delta t$ as Δt approaches zero is called the **derivative** of *x* with respect to *t* and is written dx/dt . We use the symbol v_x , with no "av" subscript, for the instantaneous velocity along the *x*-axis, or the **instantaneous *x*-velocity**:

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (2.3)$$

The instantaneous *x*-velocity of a particle in straight-line motion ...
 ... equals the limit of the particle's average *x*-velocity as the time interval approaches zero ...
 ... and equals the instantaneous rate of change of the particle's *x*-coordinate.

The time interval Δt is always positive, so v_x has the same algebraic sign as Δx . A positive value of v_x means that *x* is increasing and the motion is in the positive *x*-direction; a negative value of v_x means that *x* is decreasing and the motion is in the negative *x*-direction. An object can have positive *x* and negative v_x , or the reverse; *x* tells us where the object is, while v_x tells us how it's moving (Fig. 2.5). The rules that we presented in Table 2.1 (Section 2.1) for the sign of average *x*-velocity $v_{\text{av-}x}$ also apply to the sign of instantaneous *x*-velocity v_x .

Instantaneous velocity, like average velocity, is a vector; Eq. (2.3) defines its *x*-component. In straight-line motion, all other components of instantaneous velocity are zero. In this case we often call v_x simply the instantaneous velocity. (In Chapter 3 we'll deal with the general case in which the instantaneous velocity can have nonzero *x*-, *y*-, and *z*-components.) When we use the term "velocity," we'll always mean instantaneous rather than average velocity.

Figure 2.4 The winner of a 50 m swimming race is the swimmer whose average velocity has the greatest magnitude—that is, the swimmer who traverses a displacement Δx of 50 m in the shortest elapsed time Δt .

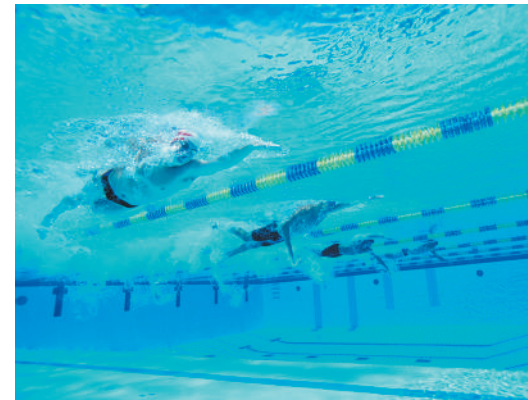


Figure 2.5 In any problem involving straight-line motion, the choice of which direction is positive and which is negative is entirely up to you.

... A bicyclist moving to the left ...



... has a negative *x*-velocity v_x if we choose the positive *x*-direction to the right ...

... but has a positive *x*-velocity v_x if we choose the positive *x*-direction to the left.

“Velocity” and “speed” are used interchangeably in everyday language, but they have distinct definitions in physics. We use the term **speed** to denote distance traveled divided by time, on either an average or an instantaneous basis. Instantaneous *speed*, for which we use the symbol v with *no* subscripts, measures how fast a particle is moving; instantaneous *velocity* measures how fast *and* in what direction it’s moving. Instantaneous speed is the magnitude of instantaneous velocity and so can never be negative. For example, a particle with instantaneous velocity $v_x = 25$ m/s and a second particle with $v_x = -25$ m/s are moving in opposite directions at the same instantaneous speed 25 m/s.

CAUTION Average speed and average velocity Average speed is *not* the magnitude of average velocity. When César Cielo set a world record in 2009 by swimming 100.0 m in 46.91 s, his average speed was $(100.0 \text{ m})/(46.91 \text{ s}) = 2.132$ m/s. But because he swam two lengths in a 50 m pool, he started and ended at the same point and so had zero total displacement and zero average *velocity*! Both average speed and instantaneous speed are scalars, not vectors, because these quantities contain no information about direction. **I**

EXAMPLE 2.1 Average and instantaneous velocities

A cheetah is crouched 20 m to the east of a vehicle (**Fig. 2.6a**). At time $t = 0$ the cheetah begins to run due east toward an antelope that is 50 m to the east of the vehicle. During the first 2.0 s of the chase, the cheetah’s x -coordinate varies with time according to the equation $x = 20 \text{ m} + (5.0 \text{ m/s}^2)t^2$. (a) Find the cheetah’s displacement between $t_1 = 1.0$ s and $t_2 = 2.0$ s. (b) Find its average velocity during that interval. (c) Find its instantaneous velocity at $t_1 = 1.0$ s by taking $\Delta t = 0.1$ s, then 0.01 s, then 0.001 s. (d) Derive an expression for the cheetah’s instantaneous velocity as a function of time, and use it to find v_x at $t = 1.0$ s and $t = 2.0$ s.

IDENTIFY and SET UP Figure 2.6b shows our sketch of the cheetah’s motion. We use Eq. (2.1) for displacement, Eq. (2.2) for average velocity, and Eq. (2.3) for instantaneous velocity.

EXECUTE (a) At $t_1 = 1.0$ s and $t_2 = 2.0$ s the cheetah’s positions x_1 and x_2 are

$$\begin{aligned}x_1 &= 20 \text{ m} + (5.0 \text{ m/s}^2)(1.0 \text{ s})^2 = 25 \text{ m} \\x_2 &= 20 \text{ m} + (5.0 \text{ m/s}^2)(2.0 \text{ s})^2 = 40 \text{ m}\end{aligned}$$

The displacement during this 1.0 s interval is

$$\Delta x = x_2 - x_1 = 40 \text{ m} - 25 \text{ m} = 15 \text{ m}$$

(b) The average x -velocity during this interval is

$$v_{\text{av-}x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{40 \text{ m} - 25 \text{ m}}{2.0 \text{ s} - 1.0 \text{ s}} = \frac{15 \text{ m}}{1.0 \text{ s}} = 15 \text{ m/s}$$

(c) With $\Delta t = 0.1$ s the time interval is from $t_1 = 1.0$ s to a new $t_2 = 1.1$ s. At t_2 the position is

$$x_2 = 20 \text{ m} + (5.0 \text{ m/s}^2)(1.1 \text{ s})^2 = 26.05 \text{ m}$$

The average x -velocity during this 0.1 s interval is

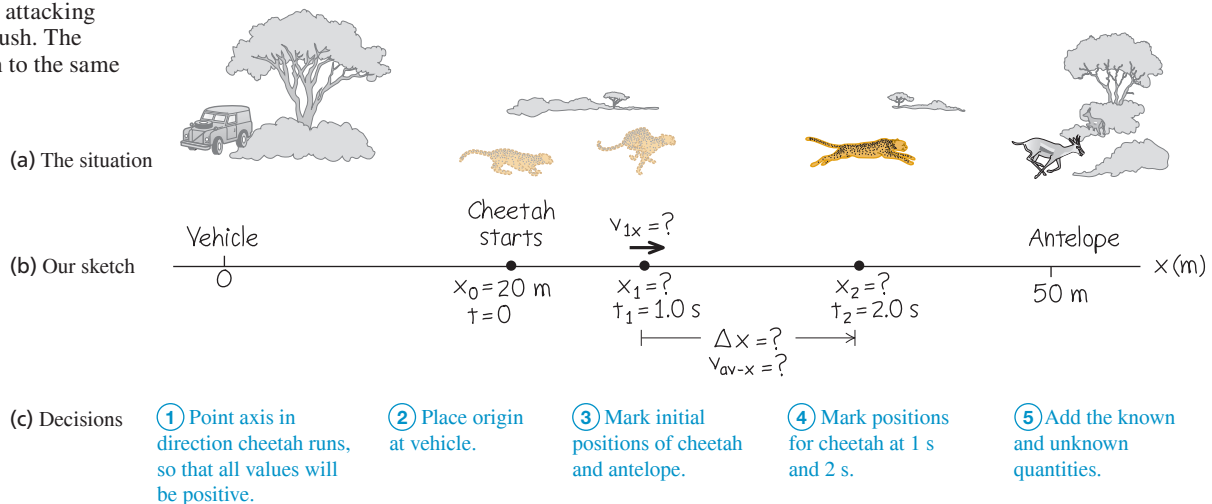
$$v_{\text{av-}x} = \frac{26.05 \text{ m} - 25 \text{ m}}{1.1 \text{ s} - 1.0 \text{ s}} = 10.5 \text{ m/s}$$

Following this pattern, you can calculate the average x -velocities for 0.01 s and 0.001 s intervals: The results are 10.05 m/s and 10.005 m/s. As Δt gets smaller, the average x -velocity gets closer to 10.0 m/s, so we conclude that the instantaneous x -velocity at $t = 1.0$ s is 10.0 m/s. (We suspended the rules for significant-figure counting in these calculations.)

(d) From Eq. (2.3) the instantaneous x -velocity is $v_x = dx/dt$. The derivative of a constant is zero and the derivative of t^2 is $2t$, so

$$\begin{aligned}v_x &= \frac{dx}{dt} = \frac{d}{dt} [20 \text{ m} + (5.0 \text{ m/s}^2)t^2] \\&= 0 + (5.0 \text{ m/s}^2)(2t) = (10 \text{ m/s}^2)t\end{aligned}$$

Figure 2.6 A cheetah attacking an antelope from ambush. The animals are not drawn to the same scale as the axis.



At $t = 1.0$ s, this yields $v_x = 10$ m/s, as we found in part (c); at $t = 2.0$ s, $v_x = 20$ m/s.

EVALUATE Our results show that the cheetah picked up speed from $t = 0$ (when it was at rest) to $t = 1.0$ s ($v_x = 10$ m/s) to $t = 2.0$ s ($v_x = 20$ m/s). This makes sense; the cheetah covered only 5 m during the interval $t = 0$ to $t = 1.0$ s, but it covered 15 m during the interval $t = 1.0$ s to $t = 2.0$ s.

KEYCONCEPT To calculate the average velocity of an object in straight-line motion, first find its displacement (final coordinate minus initial coordinate) during a time interval. Then divide by the time interval. To calculate the object's instantaneous velocity (its average velocity over an infinitesimally short time interval), take the derivative of its position with respect to time.

Finding Velocity on an x - t Graph

We can also find the x -velocity of a particle from the graph of its position as a function of time. Suppose we want to find the x -velocity of the dragster in Fig. 2.1 at point P_1 . As point P_2 in Fig. 2.1 approaches point P_1 , point p_2 in the x - t graphs of Figs. 2.7a and 2.7b approaches point p_1 and the average x -velocity is calculated over shorter time intervals Δt . In the limit that $\Delta t \rightarrow 0$, shown in Fig. 2.7c, the slope of the line p_1p_2 equals the slope of the line tangent to the curve at point p_1 . Thus, *on a graph of position as a function of time for straight-line motion, the instantaneous x -velocity at any point is equal to the slope of the tangent to the curve at that point.*

If the tangent to the x - t curve slopes upward to the right, as in Fig. 2.7c, then its slope is positive, the x -velocity is positive, and the motion is in the positive x -direction. If the tangent slopes downward to the right, the slopes of the x - t graph and the x -velocity are negative, and the motion is in the negative x -direction. When the tangent is horizontal, the slope and the x -velocity are zero. **Figure 2.8** illustrates these three possibilities.

Figure 2.7 Using an x - t graph to go from (a), (b) average x -velocity to (c) instantaneous x -velocity v_x . In (c) we find the slope of the tangent to the x - t curve by dividing any vertical interval (with distance units) along the tangent by the corresponding horizontal interval (with time units).

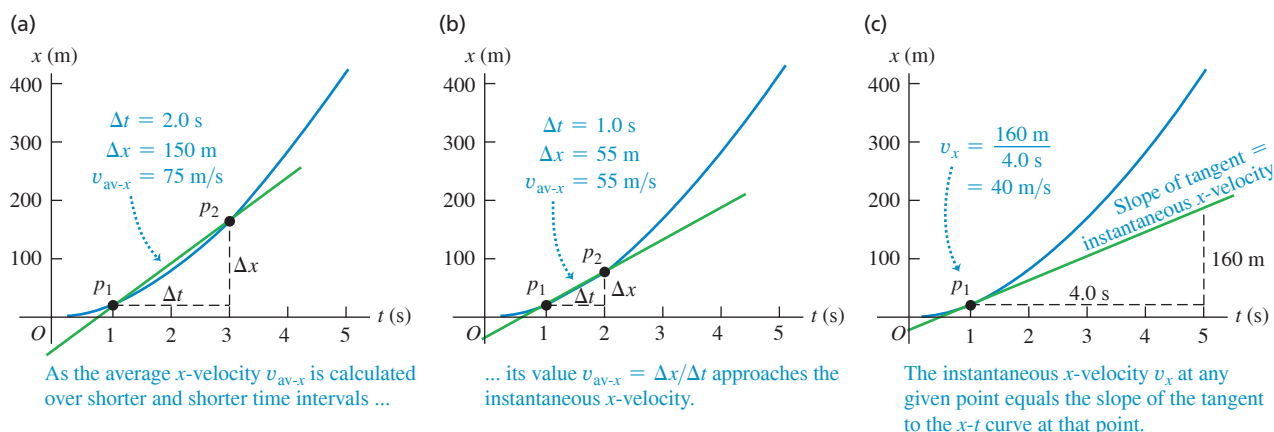
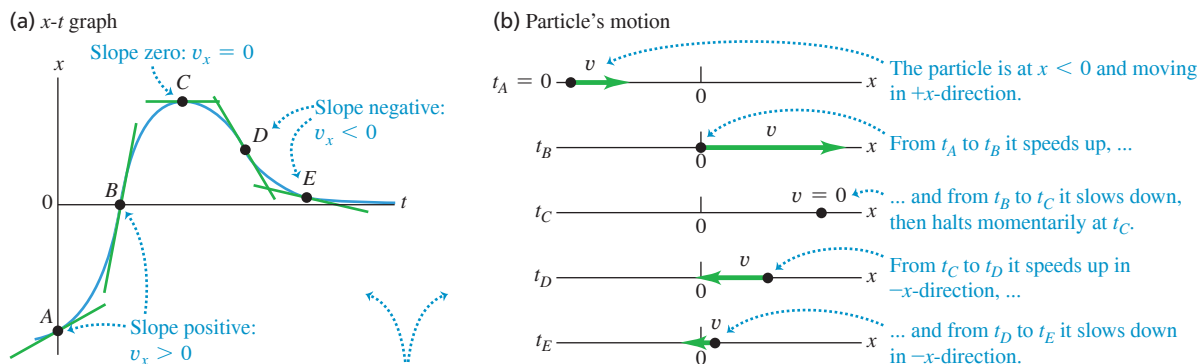


Figure 2.8 (a) The x - t graph of the motion of a particular particle. (b) A motion diagram showing the position and velocity of the particle at each of the times labeled on the x - t graph.



- On an x - t graph, the slope of the tangent at any point equals the particle's velocity at that point.
- The steeper the slope (positive or negative), the greater the particle's speed in the positive or negative x -direction.

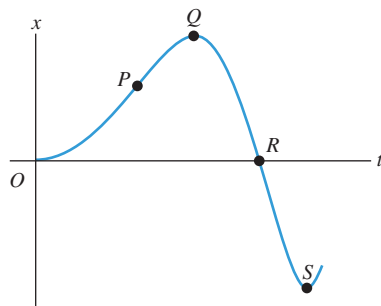
Figure 2.9 An x - t graph for a particle.

Figure 2.8 depicts the motion of a particle in two ways: as (a) an x - t graph and (b) a **motion diagram** that shows the particle's position at various instants (like frames from a video of the particle's motion) as well as arrows to represent the particle's velocity at each instant. We'll use both x - t graphs and motion diagrams in this chapter to represent motion. You'll find it helpful to draw *both* an x - t graph and a motion diagram when you solve any problem involving motion.

TEST YOUR UNDERSTANDING OF SECTION 2.2 Figure 2.9 is an x - t graph of the motion of a particle. (a) Rank the values of the particle's x -velocity v_x at points P , Q , R , and S from most positive to most negative. (b) At which points is v_x positive? (c) At which points is v_x negative? (d) At which points is v_x zero? (e) Rank the values of the particle's *speed* at points P , Q , R , and S from fastest to slowest.

ANSWER

(a) P , Q , and S (tie), R . The x -velocity is positive (b) positive when the slope of the x - t graph is positive (P), (c) negative when the slope is negative (R), and (d) zero when the slope is zero (Q and S). (e) The speed is greatest when the slope of the x - t graph is steepest (either positive or negative) and zero when the slope is zero.

2.3 AVERAGE AND INSTANTANEOUS ACCELERATION

Just as velocity describes the rate of change of position with time, *acceleration* **?** describes the rate of change of velocity with time. Like velocity, acceleration is a vector quantity. When the motion is along a straight line, its only nonzero component is along that line. In everyday language, acceleration refers only to speeding up; in physics, acceleration refers to *any* kind of velocity change, so we say an object accelerates if it is either speeding up or slowing down.

Average Acceleration

Let's consider again a particle moving along the x -axis. Suppose that at time t_1 the particle is at point P_1 and has x -component of (instantaneous) velocity v_{1x} , and at a later time t_2 it is at point P_2 and has x -component of velocity v_{2x} . So the x -component of velocity changes by an amount $\Delta v_x = v_{2x} - v_{1x}$ during the time interval $\Delta t = t_2 - t_1$. As the particle moves from P_1 to P_2 , its **average acceleration** is a vector quantity whose x -component $a_{\text{av-}x}$ (called the **average x -acceleration**) equals Δv_x , the change in the x -component of velocity, divided by the time interval Δt :

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} \quad (2.4)$$

Average x -acceleration of a particle in straight-line motion during time interval from t_1 to t_2
 Change in x -component of the particle's velocity
 Final x -velocity minus initial x -velocity
Time interval Final time minus initial time

For straight-line motion along the x -axis we'll often call $a_{\text{av-}x}$ simply the average acceleration. (We'll encounter the other components of the average acceleration vector in Chapter 3.)

If we express velocity in meters per second and time in seconds, then average acceleration is in meters per second per second. This is usually written as m/s^2 and is read "meters per second squared."

CAUTION

Don't confuse velocity and acceleration Velocity describes how an object's position changes with time; it tells us how fast and in what direction the object moves. Acceleration describes how the velocity changes with time; it tells us how the speed and direction of motion change. Another difference is that you can *feel* acceleration but you can't feel velocity. If you're a passenger in a car that accelerates forward and gains speed, you feel pushed backward in your seat; if it accelerates backward and loses speed, you feel pushed forward. If the velocity is constant and there's no acceleration, you feel neither sensation. (We'll explain these sensations in Chapter 4.)

EXAMPLE 2.2 Average acceleration

An astronaut has left an orbiting spacecraft to test a new personal maneuvering unit. As she moves along a straight line, her partner on the spacecraft measures her velocity every 2.0 s, starting at time $t = 1.0$ s:

t	v_x	t	v_x
1.0 s	0.8 m/s	9.0 s	-0.4 m/s
3.0 s	1.2 m/s	11.0 s	-1.0 m/s
5.0 s	1.6 m/s	13.0 s	-1.6 m/s
7.0 s	1.2 m/s	15.0 s	-0.8 m/s

Find the average x -acceleration, and state whether the speed of the astronaut increases or decreases over each of these 2.0 s time intervals: (a) $t_1 = 1.0$ s to $t_2 = 3.0$ s; (b) $t_1 = 5.0$ s to $t_2 = 7.0$ s; (c) $t_1 = 9.0$ s to $t_2 = 11.0$ s; (d) $t_1 = 13.0$ s to $t_2 = 15.0$ s.

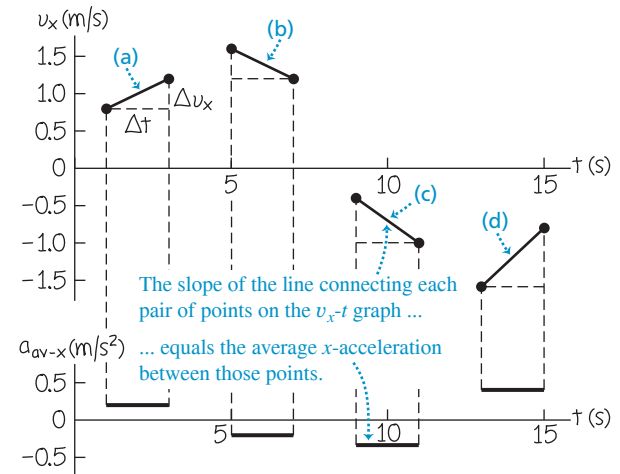
IDENTIFY and SET UP We'll use Eq. (2.4) to determine the average acceleration a_{av-x} from the change in velocity over each time interval. To find the changes in speed, we'll use the idea that speed v is the magnitude of the instantaneous velocity v_x .

The upper part of **Fig. 2.10** is our graph of the x -velocity as a function of time. On this v_x - t graph, the slope of the line connecting the endpoints of each interval is the average x -acceleration $a_{av-x} = \Delta v_x / \Delta t$ for that interval. The four slopes (and thus the *signs* of the average accelerations) are, from left to right, positive, negative, negative, and positive. The third and fourth slopes (and thus the average accelerations themselves) have greater magnitude than the first and second.

EXECUTE Using Eq. (2.4), we find:

- (a) $a_{av-x} = (1.2 \text{ m/s} - 0.8 \text{ m/s}) / (3.0 \text{ s} - 1.0 \text{ s}) = 0.2 \text{ m/s}^2$. The speed (magnitude of instantaneous x -velocity) increases from 0.8 m/s to 1.2 m/s.
- (b) $a_{av-x} = (1.2 \text{ m/s} - 1.6 \text{ m/s}) / (7.0 \text{ s} - 5.0 \text{ s}) = -0.2 \text{ m/s}^2$. The speed decreases from 1.6 m/s to 1.2 m/s.
- (c) $a_{av-x} = [-1.0 \text{ m/s} - (-0.4 \text{ m/s})] / (11.0 \text{ s} - 9.0 \text{ s}) = -0.3 \text{ m/s}^2$. The speed increases from 0.4 m/s to 1.0 m/s.
- (d) $a_{av-x} = [-0.8 \text{ m/s} - (-1.6 \text{ m/s})] / (15.0 \text{ s} - 13.0 \text{ s}) = 0.4 \text{ m/s}^2$. The speed decreases from 1.6 m/s to 0.8 m/s.

Figure 2.10 Our graphs of x -velocity versus time (top) and average x -acceleration versus time (bottom) for the astronaut.



In the lower part of Fig. 2.10, we graph the values of a_{av-x} .

EVALUATE The signs and relative magnitudes of the average accelerations agree with our qualitative predictions.

Notice that when the average x -acceleration has the *same* algebraic sign as the initial velocity, as in intervals (a) and (c), the astronaut goes faster. When a_{av-x} has the *opposite* algebraic sign from the initial velocity, as in intervals (b) and (d), she slows down. Thus positive x -acceleration means speeding up if the x -velocity is positive [interval (a)] but slowing down if the x -velocity is negative [interval (d)]. Similarly, negative x -acceleration means speeding up if the x -velocity is negative [interval (c)] but slowing down if the x -velocity is positive [interval (b)].

KEYCONCEPT To calculate the average acceleration of an object in straight-line motion, first find the change in its velocity (final velocity minus initial velocity) during a time interval. Then divide by the time interval.

Instantaneous Acceleration

We can now define **instantaneous acceleration** by following the same procedure that we used to define instantaneous velocity. Suppose a race car driver is driving along a straightaway as shown in **Fig. 2.11**. To define the instantaneous acceleration at point P_1 , we take point P_2 in Fig. 2.11 to be closer and closer to P_1 so that the average acceleration is computed over shorter and shorter time intervals. Thus

The **instantaneous x -acceleration** of a particle in straight-line motion ...

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} \quad (2.5)$$

... equals the limit of the particle's average x -acceleration as the time interval approaches zero ...

... and equals the instantaneous rate of change of the particle's x -velocity.

Figure 2.11 A Grand Prix car at two points on the straightaway.



In Eq. (2.5) a_x is the x -component of the acceleration vector, which we call the **instantaneous x -acceleration**; in straight-line motion, all other components of this vector are zero. From now on, when we use the term “acceleration,” we’ll always mean instantaneous acceleration, not average acceleration.

EXAMPLE 2.3 Average and instantaneous accelerations

Suppose the x -velocity v_x of the car in Fig. 2.11 at any time t is given by the equation

$$v_x = 60 \text{ m/s} + (0.50 \text{ m/s}^3)t^2$$

(a) Find the change in x -velocity of the car in the time interval $t_1 = 1.0 \text{ s}$ to $t_2 = 3.0 \text{ s}$. (b) Find the average x -acceleration in this time interval. (c) Find the instantaneous x -acceleration at time $t_1 = 1.0 \text{ s}$ by taking Δt to be first 0.1 s , then 0.01 s , then 0.001 s . (d) Derive an expression for the instantaneous x -acceleration as a function of time, and use it to find a_x at $t = 1.0 \text{ s}$ and $t = 3.0 \text{ s}$.

IDENTIFY and SET UP This example is analogous to Example 2.1 in Section 2.2. In that example we found the average x -velocity from the change in position over shorter and shorter time intervals, and we obtained an expression for the instantaneous x -velocity by differentiating the position as a function of time. In this example we have an exact parallel. Using Eq. (2.4), we’ll find the average x -acceleration from the change in x -velocity over a time interval. Likewise, using Eq. (2.5), we’ll obtain an expression for the instantaneous x -acceleration by differentiating the x -velocity as a function of time.

EXECUTE (a) Before we can apply Eq. (2.4), we must find the x -velocity at each time from the given equation. At $t_1 = 1.0 \text{ s}$ and $t_2 = 3.0 \text{ s}$, the velocities are

$$v_{1x} = 60 \text{ m/s} + (0.50 \text{ m/s}^3)(1.0 \text{ s})^2 = 60.5 \text{ m/s}$$

$$v_{2x} = 60 \text{ m/s} + (0.50 \text{ m/s}^3)(3.0 \text{ s})^2 = 64.5 \text{ m/s}$$

The change in x -velocity Δv_x between $t_1 = 1.0 \text{ s}$ and $t_2 = 3.0 \text{ s}$ is

$$\Delta v_x = v_{2x} - v_{1x} = 64.5 \text{ m/s} - 60.5 \text{ m/s} = 4.0 \text{ m/s}$$

(b) The average x -acceleration during this time interval of duration $t_2 - t_1 = 2.0 \text{ s}$ is

$$a_{\text{av-}x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{4.0 \text{ m/s}}{2.0 \text{ s}} = 2.0 \text{ m/s}^2$$

During this time interval the x -velocity and average x -acceleration have the same algebraic sign (in this case, positive), and the car speeds up.

(c) When $\Delta t = 0.1 \text{ s}$, we have $t_2 = 1.1 \text{ s}$. Proceeding as before, we find

$$v_{2x} = 60 \text{ m/s} + (0.50 \text{ m/s}^3)(1.1 \text{ s})^2 = 60.605 \text{ m/s}$$

$$\Delta v_x = 0.105 \text{ m/s}$$

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{0.105 \text{ m/s}}{0.1 \text{ s}} = 1.05 \text{ m/s}^2$$

You should follow this pattern to calculate $a_{\text{av-}x}$ for $\Delta t = 0.01 \text{ s}$ and $\Delta t = 0.001 \text{ s}$; the results are $a_{\text{av-}x} = 1.005 \text{ m/s}^2$ and $a_{\text{av-}x} = 1.0005 \text{ m/s}^2$, respectively. As Δt gets smaller, the average x -acceleration gets closer to 1.0 m/s^2 , so the instantaneous x -acceleration at $t = 1.0 \text{ s}$ is 1.0 m/s^2 .

(d) By Eq. (2.5) the instantaneous x -acceleration is $a_x = dv_x/dt$. The derivative of a constant is zero and the derivative of t^2 is $2t$, so

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}[60 \text{ m/s} + (0.50 \text{ m/s}^3)t^2] = (0.50 \text{ m/s}^3)(2t) = (1.0 \text{ m/s}^3)t$$

When $t = 1.0 \text{ s}$,

$$a_x = (1.0 \text{ m/s}^3)(1.0 \text{ s}) = 1.0 \text{ m/s}^2$$

When $t = 3.0 \text{ s}$,

$$a_x = (1.0 \text{ m/s}^3)(3.0 \text{ s}) = 3.0 \text{ m/s}^2$$

EVALUATE Neither of the values we found in part (d) is equal to the average x -acceleration found in part (b). That’s because the car’s instantaneous x -acceleration varies with time. The rate of change of acceleration with time is sometimes called the “jerk.”

KEYCONCEPT To calculate an object’s instantaneous acceleration (its average acceleration over an infinitesimally short time interval), take the derivative of its velocity with respect to time.

Finding Acceleration on a v_x - t Graph or an x - t Graph

In Section 2.2 we interpreted average and instantaneous x -velocity in terms of the slope of a graph of position versus time. In the same way, we can interpret average and instantaneous x -acceleration by using a graph of instantaneous velocity v_x versus time t —that is, a **v_x - t graph** (Fig. 2.12). Points p_1 and p_2 on the graph correspond to points P_1 and P_2 in Fig. 2.11. The average x -acceleration $a_{\text{av-}x} = \Delta v_x / \Delta t$ during this interval is the slope of the line p_1p_2 .

As point P_2 in Fig. 2.11 approaches point P_1 , point p_2 in the v_x - t graph of Fig. 2.12 approaches point p_1 , and the slope of the line p_1p_2 approaches the slope of the line tangent to the curve at point p_1 . Thus, *on a graph of x -velocity as a function of time, the instantaneous x -acceleration at any point is equal to the slope of the tangent to the curve at that point.* Tangents drawn at different points along the curve in Fig. 2.12 have different slopes, so the instantaneous x -acceleration varies with time.

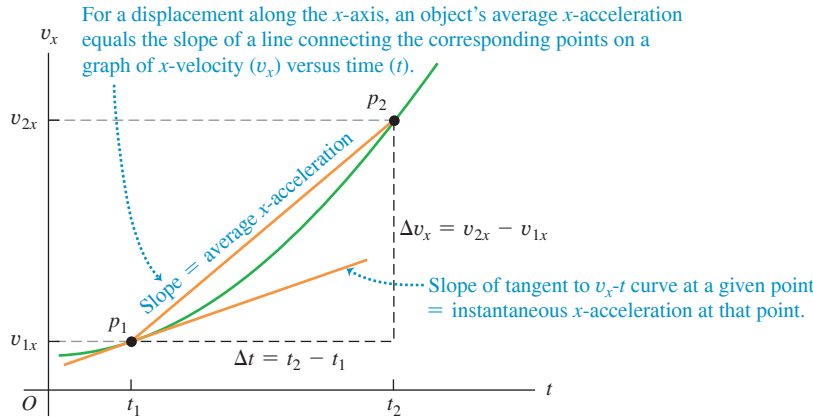


Figure 2.12 A v_x - t graph of the motion in Fig. 2.11.

CAUTION Signs of x -acceleration and x -velocity The algebraic sign of the x -acceleration does not tell you whether an object is speeding up or slowing down. You must compare the signs of the x -velocity and the x -acceleration. As we saw in Example 2.2, when v_x and a_x have the *same* sign, the object is speeding up. When v_x and a_x have *opposite* signs, the object is slowing down. Table 2.3 summarizes these rules, and Fig. 2.13 illustrates some of them.

The term “deceleration” is sometimes used for a decrease in speed. Because it may mean positive or negative a_x , depending on the sign of v_x , we avoid this term.

We can also learn about the acceleration of an object from a graph of its *position* versus time. Because $a_x = dv_x/dt$ and $v_x = dx/dt$, we can write

$$a_x = \frac{dv_x}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2x}{dt^2} \quad (2.6)$$

That is, a_x is the second derivative of x with respect to t . The second derivative of any function is directly related to the *concavity* or *curvature* of the graph of that function (Fig. 2.14, next page). At a point where the x - t graph is concave up (curved upward), such as point A or E in Fig. 2.14a, the x -acceleration is positive and v_x is increasing. At a point where the x - t graph is concave down (curved downward), such as point C in Fig. 2.14a, the x -acceleration is negative and v_x is decreasing. At a point where the x - t graph has no curvature, such as the inflection points B and D in Fig. 2.14a, the x -acceleration is zero and the velocity is not changing.

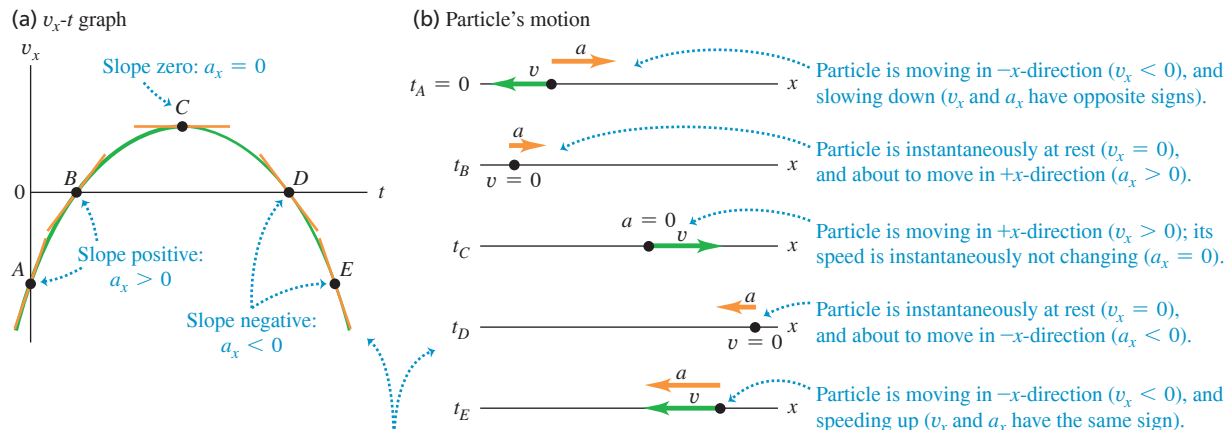
Examining the curvature of an x - t graph is an easy way to identify the *sign* of acceleration. This technique is less helpful for determining numerical values of acceleration because the curvature of a graph is hard to measure accurately.

TABLE 2.3 Rules for the Sign of x -Acceleration

If x -velocity is:	... x -acceleration is:
Positive & increasing (getting more positive)	Positive: Particle is moving in $+x$ -direction & speeding up
Positive & decreasing (getting less positive)	Negative: Particle is moving in $+x$ -direction & slowing down
Negative & increasing (getting less negative)	Positive: Particle is moving in $-x$ -direction & slowing down
Negative & decreasing (getting more negative)	Negative: Particle is moving in $-x$ -direction & speeding up

Note: These rules apply to both the average x -acceleration a_{av-x} and the instantaneous x -acceleration a_x .

Figure 2.13 (a) The v_x - t graph of the motion of a different particle from that shown in Fig. 2.8. (b) A motion diagram showing the position, velocity, and acceleration of the particle at each of the times labeled on the v_x - t graph.



- On a v_x - t graph, the slope of the tangent at any point equals the particle's acceleration at that point.
- The steeper the slope (positive or negative), the greater the particle's acceleration in the positive or negative x -direction.

Figure 2.14 (a) The same x - t graph as shown in Fig. 2.8a. (b) A motion diagram showing the position, velocity, and acceleration of the particle at each of the times labeled on the x - t graph.

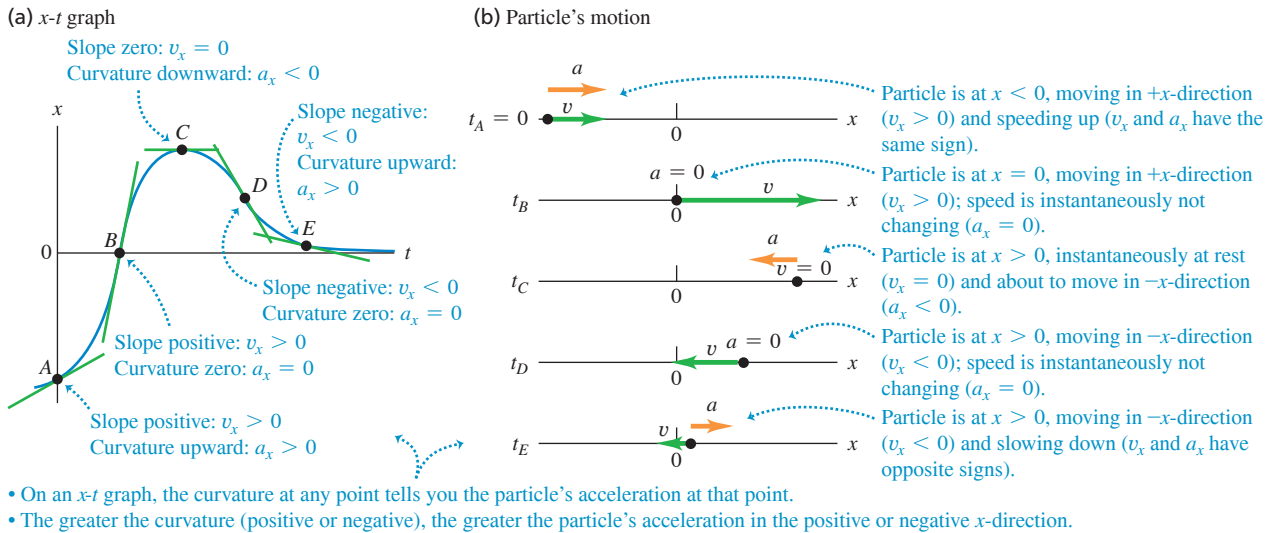


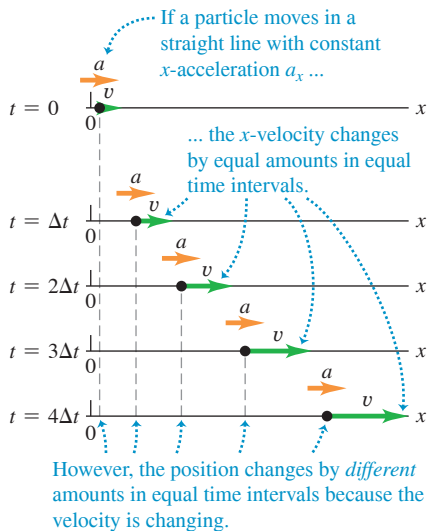
Table 2.4 summarizes what you can learn from the x - t graph and v_x - t graph of the straight-line motion of a particle.

TABLE 2.4 What x - t Graphs and v_x - t Graphs Tell You

	On an x - t graph	On a v_x - t graph
The <i>value</i> of the graph at a given time tells you...	The coordinate x at that time	The velocity v_x at that time
The <i>slope</i> of the graph at a given time tells you...	The velocity v_x at that time	The acceleration a_x at that time
The <i>concavity</i> or <i>curvature</i> of the graph at a given time tells you...	The acceleration a_x at that time	If the acceleration a_x is changing at that time

TEST YOUR UNDERSTANDING OF SECTION 2.3 Look again at the x - t graph in Fig. 2.9 at the end of Section 2.2. (a) At which of the points P , Q , R , and S is the x -acceleration a_x positive? (b) At which points is the x -acceleration negative? (c) At which points does the x -acceleration appear to be zero? (d) At each point state whether the velocity is increasing, decreasing, or not changing.

Figure 2.15 A motion diagram for a particle moving in a straight line in the positive x -direction with constant positive x -acceleration a_x .



ANSWER (a) S , where the x - t graph is curved upward (concave up). (b) Q , where the x - t graph is curved downward (concave down). (c) P and R , where the x - t graph is not curved either up or down. (d) At P , $a_x = 0$ (velocity is *not changing*; i.e., changing from positive to zero to negative); at R , $a_x = 0$ (velocity is *not changing*; i.e., changing from negative to zero to positive). (velocity is *increasing*, i.e., changing from negative to zero to positive).

2.4 MOTION WITH CONSTANT ACCELERATION

The simplest kind of accelerated motion is straight-line motion with *constant* acceleration. In this case the velocity changes at the same rate throughout the motion. As an example, a falling object has a constant acceleration if the effects of the air are not important. The same is true for an object sliding on an incline or along a rough horizontal surface, or for an airplane being catapulted from the deck of an aircraft carrier.

Figure 2.15 is a motion diagram showing the position, velocity, and acceleration of a particle moving with constant acceleration. Figures 2.16 and 2.17 depict this same motion in the form of graphs. Since the x -acceleration is constant, the a_x - t graph (graph of x -acceleration versus time) in Fig. 2.16 is a horizontal line. The graph of x -velocity versus time, or v_x - t graph, has a constant *slope* because the acceleration is constant, so this graph is a straight line (Fig. 2.17).

When the x -acceleration a_x is constant, the average x -acceleration a_{av-x} for any time interval is the same as a_x . This makes it easy to derive equations for the position x and the x -velocity v_x as functions of time. To find an equation for v_x , we first replace a_{av-x} in Eq. (2.4) by a_x :

$$a_x = \frac{v_{2x} - v_{1x}}{t_2 - t_1} \quad (2.7)$$

Now we let $t_1 = 0$ and let t_2 be any later time t . We use the symbol v_{0x} for the *initial* x -velocity at time $t = 0$; the x -velocity at the later time t is v_x . Then Eq. (2.7) becomes

$$a_x = \frac{v_x - v_{0x}}{t - 0} \quad \text{or}$$

$$v_x = v_{0x} + a_x t \quad (2.8)$$

$\xrightarrow{\text{x-velocity at time } t \text{ of a particle with constant x-acceleration}}$ $\xrightarrow{\text{x-velocity of the particle at time 0}}$ $\xrightarrow{\text{Constant x-acceleration of the particle}}$ $\xrightarrow{\text{Time}}$

In Eq. (2.8) the term $a_x t$ is the product of the constant rate of change of x -velocity, a_x , and the time interval t . Therefore it equals the *total* change in x -velocity from $t = 0$ to time t . The x -velocity v_x at any time t then equals the initial x -velocity v_{0x} (at $t = 0$) plus the change in x -velocity $a_x t$ (Fig. 2.17).

Equation (2.8) also says that the change in x -velocity $v_x - v_{0x}$ of the particle between $t = 0$ and any later time t equals the *area* under the a_x - t graph between those two times. You can verify this from Fig. 2.16: Under this graph is a rectangle of vertical side a_x , horizontal side t , and area $a_x t$. From Eq. (2.8) the area $a_x t$ is indeed equal to the change in velocity $v_x - v_{0x}$. In Section 2.6 we'll show that even if the x -acceleration is not constant, the change in x -velocity during a time interval is still equal to the area under the a_x - t curve, although then Eq. (2.8) does not apply.

Next we'll derive an equation for the position x as a function of time when the x -acceleration is constant. To do this, we use two different expressions for the average x -velocity v_{av-x} during the interval from $t = 0$ to any later time t . The first expression comes from the definition of v_{av-x} , Eq. (2.2), which is true whether or not the acceleration is constant. The position at time $t = 0$, called the *initial position*, is x_0 . The position at time t is simply x . Thus for the time interval $\Delta t = t - 0$ the displacement is $\Delta x = x - x_0$, and Eq. (2.2) gives

$$v_{av-x} = \frac{x - x_0}{t} \quad (2.9)$$

To find a second expression for v_{av-x} , note that the x -velocity changes at a constant rate if the x -acceleration is constant. In this case the average x -velocity for the time interval from 0 to t is simply the average of the x -velocities at the beginning and end of the interval:

$$v_{av-x} = \frac{1}{2}(v_{0x} + v_x) \quad (\text{constant } x\text{-acceleration only}) \quad (2.10)$$

[Equation (2.10) is *not* true if the x -acceleration varies during the time interval.] We also know that with constant x -acceleration, the x -velocity v_x at any time t is given by Eq. (2.8). Substituting that expression for v_x into Eq. (2.10), we find

$$\begin{aligned} v_{av-x} &= \frac{1}{2}(v_{0x} + v_{0x} + a_x t) \\ &= v_{0x} + \frac{1}{2}a_x t \end{aligned} \quad (\text{constant } x\text{-acceleration only}) \quad (2.11)$$

Finally, we set Eqs. (2.9) and (2.11) equal to each other and simplify:

$$v_{0x} + \frac{1}{2}a_x t = \frac{x - x_0}{t} \quad \text{or}$$

Figure 2.16 An acceleration-time (a_x - t) graph of straight-line motion with constant positive x -acceleration a_x .

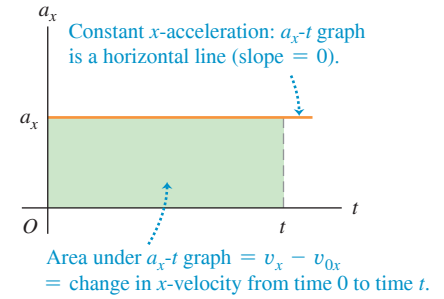
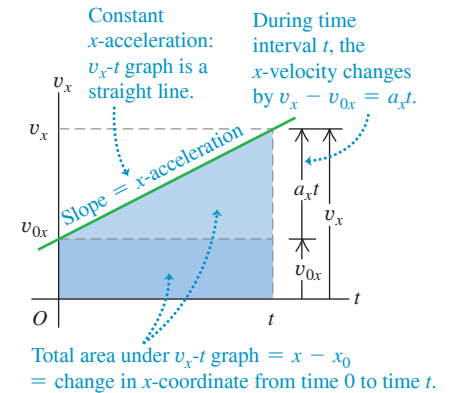


Figure 2.17 A velocity-time (v_x - t) graph of straight-line motion with constant positive x -acceleration a_x . The initial x -velocity v_{0x} is also positive in this case.



BIO APPLICATION Testing Humans

at High Accelerations In experiments carried out by the U.S. Air Force in the 1940s and 1950s, humans riding a rocket sled could withstand accelerations as great as 440 m/s^2 . The first three photos in this sequence show Air Force physician John Stapp speeding up from rest to 188 m/s ($678 \text{ km/h} = 421 \text{ mi/h}$) in just 5 s. Photos 4–6 show the even greater magnitude of acceleration as the rocket sled braked to a halt.



$$x = x_0 + v_{0x}t + \frac{1}{2}a_xt^2 \quad (2.12)$$

Position of the particle at time 0
Position at time t of a particle with constant x -acceleration
 x -velocity of the particle at time 0
Constant x -acceleration of the particle
Time

Equation (2.12) tells us that the particle's position at time t is the sum of three terms: its initial position at $t = 0$, x_0 , plus the displacement $v_{0x}t$ it would have if its x -velocity remained equal to its initial value, plus an additional displacement $\frac{1}{2}a_xt^2$ caused by the change in x -velocity.

A graph of Eq. (2.12)—that is, an x - t graph for motion with constant x -acceleration (**Fig. 2.18a**)—is always a *parabola*. Figure 2.18b shows such a graph. The curve intercepts the vertical axis (x -axis) at x_0 , the position at $t = 0$. The slope of the tangent at $t = 0$ equals v_{0x} , the initial x -velocity, and the slope of the tangent at any time t equals the x -velocity v_x at that time. The slope and x -velocity are continuously increasing, so the x -acceleration a_x is positive and the graph in Fig. 2.18b is concave up (it curves upward). If a_x is negative, the x - t graph is a parabola that is concave down (has a downward curvature).

If there is zero x -acceleration, the x - t graph is a straight line; if there is a constant x -acceleration, the additional $\frac{1}{2}a_xt^2$ term in Eq. (2.12) for x as a function of t curves the graph into a parabola (**Fig. 2.19a**). Similarly, if there is zero x -acceleration, the v_x - t graph is a horizontal line (the x -velocity is constant). Adding a constant x -acceleration in Eq. (2.8) gives a slope to the graph (Fig. 2.19b).

Here's another way to derive Eq. (2.12). Just as the change in x -velocity of the particle equals the area under the a_x - t graph, the displacement (change in position) equals the area under the v_x - t graph. So the displacement $x - x_0$ of the particle between $t = 0$ and any later time t equals the area under the v_x - t graph between those times. In Fig. 2.17 we divide the area under the graph into a dark-colored rectangle (vertical side v_{0x} , horizontal side t , and area $v_{0x}t$) and a light-colored right triangle (vertical side a_xt , horizontal side t , and area $\frac{1}{2}(a_xt)(t) = \frac{1}{2}a_xt^2$). The total area under the v_x - t graph is $x - x_0 = v_{0x}t + \frac{1}{2}a_xt^2$, in accord with Eq. (2.12).

It's often useful to have a relationship for position, x -velocity, and (constant) x -acceleration that does not involve time. To obtain this, we first solve Eq. (2.8) for t and then substitute the resulting expression into Eq. (2.12):

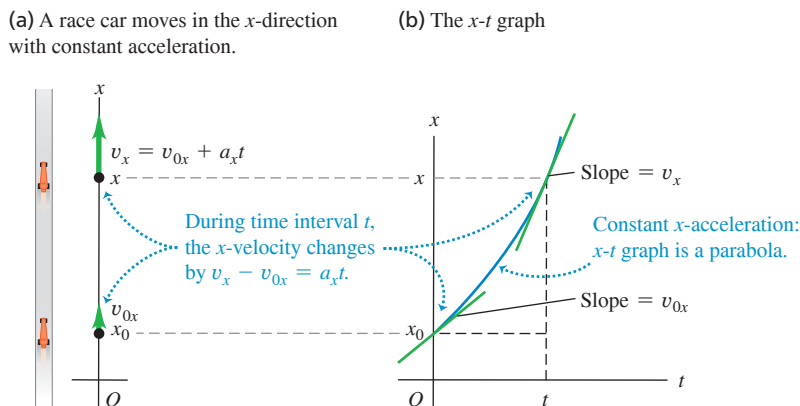
$$t = \frac{v_x - v_{0x}}{a_x}$$

$$x = x_0 + v_{0x}\left(\frac{v_x - v_{0x}}{a_x}\right) + \frac{1}{2}a_x\left(\frac{v_x - v_{0x}}{a_x}\right)^2$$

We transfer the term x_0 to the left side, multiply through by $2a_x$, and simplify:

$$2a_x(x - x_0) = 2v_{0x}v_x - 2v_{0x}^2 + v_x^2 - 2v_{0x}v_x + v_{0x}^2$$

Figure 2.18 (a) Straight-line motion with constant acceleration. (b) A position-time (x - t) graph for this motion (the same motion as is shown in Figs. 2.15, 2.16, and 2.17). For this motion the initial position x_0 , the initial velocity v_{0x} , and the acceleration a_x are all positive.



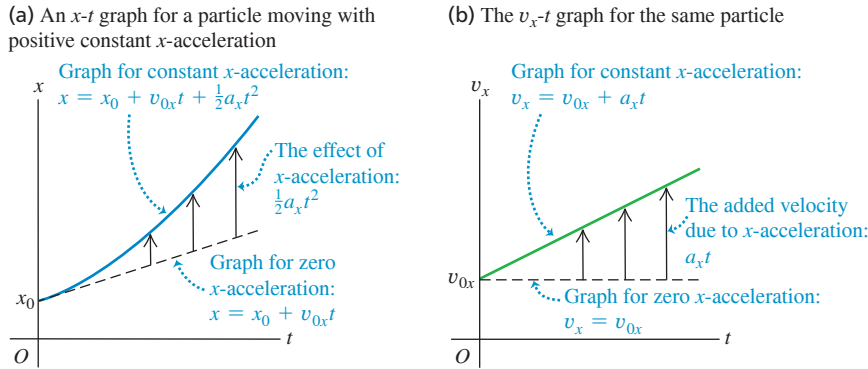


Figure 2.19 (a) How a constant x -acceleration affects a particle's (a) x - t graph and (b) v_x - t graph.

Finally,

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0) \quad (2.13)$$

Labels for Eq. (2.13):
 v_x^2 : x-velocity at time t of a particle with constant x-acceleration
 v_{0x}^2 : x-velocity of the particle at time 0
 $2a_x$: Constant x-acceleration of the particle
 $(x - x_0)$: Position of the particle at time t minus Position of the particle at time 0

We can get one more useful relationship by equating the two expressions for v_{av-x} , Eqs. (2.9) and (2.10), and multiplying through by t :

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t \quad (2.14)$$

Labels for Eq. (2.14):
 $x - x_0$: Position at time t of a particle with constant x-acceleration minus Position of the particle at time 0
 $\frac{1}{2}(v_{0x} + v_x)t$: Time multiplied by the average of x-velocity of the particle at time 0 and x-velocity of the particle at time t

Note that Eq. (2.14) does not contain the x -acceleration a_x . This equation can be handy when a_x is constant but its value is unknown.

Equations (2.8), (2.12), (2.13), and (2.14) are the *equations of motion with constant acceleration* (**Table 2.5**). By using these equations, we can solve *any* problem involving straight-line motion of a particle with constant acceleration.

For the particular case of motion with constant x -acceleration depicted in Fig. 2.15 and graphed in Figs. 2.16, 2.17, and 2.18, the values of x_0 , v_{0x} , and a_x are all positive. We recommend that you redraw these figures for cases in which one, two, or all three of these quantities are negative.

TABLE 2.5 Equations of Motion with Constant Acceleration

Equation		Includes Quantities	
$v_x = v_{0x} + a_x t$	(2.8)	t	v_x, a_x
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	(2.12)	t, x	a_x
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	(2.13)	x, v_x	a_x
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	(2.14)	t, x, v_x	

PROBLEM-SOLVING STRATEGY 2.1 Motion with Constant Acceleration

IDENTIFY the relevant concepts: In most straight-line motion problems, you can use the constant-acceleration Equations (2.8), (2.12), (2.13), and (2.14). If you encounter a situation in which the acceleration *isn't* constant, you'll need a different approach (see Section 2.6).

SET UP the problem using the following steps:

1. Read the problem carefully. Make a motion diagram showing the location of the particle at the times of interest. Decide where to place the origin of coordinates and which axis direction is positive. It's often helpful to place the particle at the origin at time $t = 0$; then $x_0 = 0$. Your choice of the positive axis direction automatically determines the positive directions for x -velocity and x -acceleration. If x is positive to the right of the origin, then v_x and a_x are also positive toward the right.

2. Identify the physical quantities (times, positions, velocities, and accelerations) that appear in Eqs. (2.8), (2.12), (2.13), and (2.14) and assign them appropriate symbols: t , x , x_0 , v_x , v_{0x} , and a_x , or symbols related to those. Translate the prose into physics: "When does the particle arrive at its highest point?" means "What is the value of t when x has its maximum value?" In Example 2.4, "Where is he when his speed is 25 m/s?" means "What is the value of x when $v_x = 25$ m/s?" Be alert for implicit information. For example, "A car sits at a stop light" usually means $v_{0x} = 0$.
3. List the quantities such as x , x_0 , v_x , v_{0x} , a_x , and t . Some of them will be known and some will be unknown. Write down the values of the known quantities, and decide which of the unknowns are the target variables. Make note of the *absence* of any of the quantities that appear in the four constant-acceleration equations.

Continued

- Use Table 2.5 to identify the applicable equations. (These are often the equations that don't include any of the absent quantities that you identified in step 3.) Usually you'll find a single equation that contains only one of the target variables. Sometimes you must find two equations, each containing the same two unknowns.
- Sketch graphs corresponding to the applicable equations. The v_x - t graph of Eq. (2.8) is a straight line with slope a_x . The x - t graph of Eq. (2.12) is a parabola that's concave up if a_x is positive and concave down if a_x is negative.

- On the basis of your experience with such problems, and taking account of what your sketched graphs tell you, make any qualitative and quantitative predictions you can about the solution.

EXECUTE *the solution:* If a single equation applies, solve it for the target variable, *using symbols only*; then substitute the known values and calculate the value of the target variable. If you have two equations in two unknowns, solve them simultaneously for the target variables.

EVALUATE *your answer:* Take a hard look at your results to see whether they make sense. Are they within the general range of values that you expected?

EXAMPLE 2.4 Constant-acceleration calculations

A motorcyclist heading east through a small town accelerates at a constant 4.0 m/s^2 after he leaves the city limits (**Fig. 2.20**). At time $t = 0$ he is 5.0 m east of the city-limits signpost while he moves east at 15 m/s . (a) Find his position and velocity at $t = 2.0 \text{ s}$. (b) Where is he when his speed is 25 m/s ?

IDENTIFY and SET UP The x -acceleration is constant, so we can use the constant-acceleration equations. We take the signpost as the origin of coordinates ($x = 0$) and choose the positive x -axis to point east (see Fig. 2.20, which is also a motion diagram). The known variables are the initial position and velocity, $x_0 = 5.0 \text{ m}$ and $v_{0x} = 15 \text{ m/s}$, and the acceleration, $a_x = 4.0 \text{ m/s}^2$. The unknown target variables in part (a) are the values of the position x and the x -velocity v_x at $t = 2.0 \text{ s}$; the target variable in part (b) is the value of x when $v_x = 25 \text{ m/s}$.

EXECUTE (a) Since we know the values of x_0 , v_{0x} , and a_x , Table 2.5 tells us that we can find the position x at $t = 2.0 \text{ s}$ by using

Eq. (2.12) and the x -velocity v_x at this time by using Eq. (2.8):

$$\begin{aligned} x &= x_0 + v_{0x}t + \frac{1}{2}a_xt^2 \\ &= 5.0 \text{ m} + (15 \text{ m/s})(2.0 \text{ s}) + \frac{1}{2}(4.0 \text{ m/s}^2)(2.0 \text{ s})^2 = 43 \text{ m} \\ v_x &= v_{0x} + a_xt \\ &= 15 \text{ m/s} + (4.0 \text{ m/s}^2)(2.0 \text{ s}) = 23 \text{ m/s} \end{aligned}$$

(b) We want to find the value of x when $v_x = 25 \text{ m/s}$, but we don't know the time when the motorcycle has this velocity. Table 2.5 tells us that we should use Eq. (2.13), which involves x , v_x , and a_x but does *not* involve t :

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

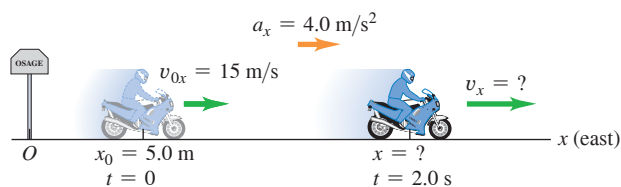
Solving for x and substituting the known values, we find

$$\begin{aligned} x &= x_0 + \frac{v_x^2 - v_{0x}^2}{2a_x} \\ &= 5.0 \text{ m} + \frac{(25 \text{ m/s})^2 - (15 \text{ m/s})^2}{2(4.0 \text{ m/s}^2)} = 55 \text{ m} \end{aligned}$$

EVALUATE You can check the result in part (b) by first using Eq. (2.8), $v_x = v_{0x} + a_xt$, to find the time at which $v_x = 25 \text{ m/s}$, which turns out to be $t = 2.5 \text{ s}$. You can then use Eq. (2.12), $x = x_0 + v_{0x}t + \frac{1}{2}a_xt^2$, to solve for x . You should find $x = 55 \text{ m}$, the same answer as above. That's the long way to solve the problem, though. The method we used in part (b) is much more efficient.

KEYCONCEPT By using one or more of the four equations in Table 2.5, you can solve any problem involving straight-line motion with constant acceleration.

Figure 2.20 A motorcyclist traveling with constant acceleration.



EXAMPLE 2.5 Two objects with different accelerations

WITH VARIATION PROBLEMS

A motorist traveling at a constant 15 m/s (54 km/h , or about 34 mi/h) passes a school crossing where the speed limit is 10 m/s (36 km/h , or about 22 mi/h). Just as the motorist passes the school-crossing sign, a police officer on a motorcycle stopped there starts in pursuit with constant acceleration 3.0 m/s^2 (**Fig. 2.21a**). (a) How much time elapses before the officer passes the motorist? At that time, (b) what is the officer's speed and (c) how far has each vehicle traveled?

IDENTIFY and SET UP Both the officer and the motorist move with constant acceleration (equal to zero for the motorist), so we can use the constant-acceleration formulas. We take the origin at the sign, so $x_0 = 0$

for both, and we take the positive direction to the right. Let x_P and x_M represent the positions of the police officer and the motorist at any time. Their initial velocities are $v_{P0x} = 0$ and $v_{M0x} = 15 \text{ m/s}$, and their accelerations are $a_{Px} = 3.0 \text{ m/s}^2$ and $a_{Mx} = 0$. Our target variable in part (a) is the time when the officer and motorist are at the same position x ; Table 2.5 tells us that Eq. (2.12) is useful for this part. In part (b) we'll use Eq. (2.8) to find the officer's speed v (the magnitude of her velocity) at the time found in part (a). In part (c) we'll use Eq. (2.12) again to find the position of either vehicle at this same time.

Figure 2.21 (a) Motion with constant acceleration overtaking motion with constant velocity. (b) A graph of x versus t for each vehicle.

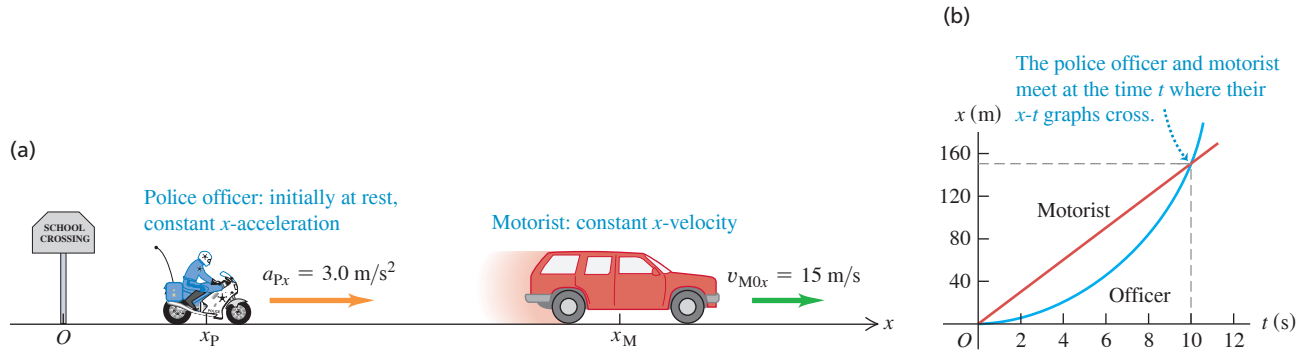


Figure 2.21b shows an x - t graph for both vehicles. The straight line represents the motorist's motion, $x_M = x_{M0} + v_{M0x}t = v_{M0x}t$. The graph for the officer's motion is the right half of a parabola with upward curvature:

$$x_P = x_{P0} + v_{P0x}t + \frac{1}{2}a_{Px}t^2 = \frac{1}{2}a_{Px}t^2$$

A good sketch shows that the officer and motorist are at the same position ($x_P = x_M$) at about $t = 10$ s, at which time both have traveled about 150 m from the sign.

EXECUTE (a) To find the value of the time t at which the motorist and police officer are at the same position, we set $x_P = x_M$ by equating the expressions above and solving that equation for t :

$$v_{M0x}t = \frac{1}{2}a_{Px}t^2 \quad \text{or} \quad t = \frac{2v_{M0x}}{a_{Px}} = \frac{2(15 \text{ m/s})}{3.0 \text{ m/s}^2} = 10 \text{ s}$$

Both vehicles have the same x -coordinate at *two* times, as Fig. 2.21b indicates. At $t = 0$ the motorist passes the officer; at $t = 10$ s the officer passes the motorist.

(b) We want the magnitude of the officer's x -velocity v_{Px} at the time t found in part (a). Substituting the values of v_{P0x} and a_{Px} into Eq. (2.8) along with $t = 10$ s from part (a), we find

$$v_{Px} = v_{P0x} + a_{Px}t = 0 + (3.0 \text{ m/s}^2)(10 \text{ s}) = 30 \text{ m/s}$$

The officer's speed is the absolute value of this, which is also 30 m/s.

(c) In 10 s the motorist travels a distance

$$x_M = v_{M0x}t = (15 \text{ m/s})(10 \text{ s}) = 150 \text{ m}$$

and the officer travels

$$x_P = \frac{1}{2}a_{Px}t^2 = \frac{1}{2}(3.0 \text{ m/s}^2)(10 \text{ s})^2 = 150 \text{ m}$$

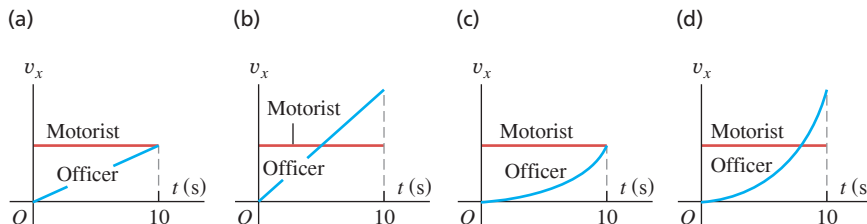
This verifies that they have gone equal distances after 10 s.

EVALUATE Our results in parts (a) and (c) agree with our estimates from our sketch. Note that when the officer passes the motorist, they do *not* have the same velocity: The motorist is moving at 15 m/s and the officer is moving at 30 m/s. You can also see this from Fig. 2.21b. Where the two x - t curves cross, their slopes (equal to the values of v_x for the two vehicles) are different.

Is it just coincidence that when the two vehicles are at the same position, the officer is going twice the speed of the motorist? Equation (2.14), $x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$, gives the answer. The motorist has constant velocity, so $v_{M0x} = v_{Mx}$, and the motorist's displacement $x - x_0$ in time t is $v_{M0x}t$. Because $v_{P0x} = 0$, in the same time t the officer's displacement is $\frac{1}{2}v_{Px}t$. The two vehicles have the same displacement in the same amount of time, so $v_{M0x}t = \frac{1}{2}v_{Px}t$ and $v_{Px} = 2v_{M0x}$ —that is, the officer has exactly twice the motorist's velocity. This is true no matter what the value of the officer's acceleration.

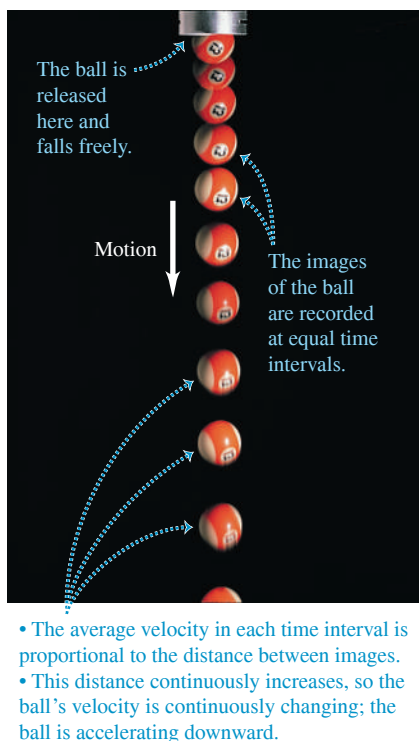
KEYCONCEPT In straight-line motion, one object meets or passes another at the time when the two objects have the same coordinate x (and so their x - t graphs cross). The objects can have different velocities at that time.

TEST YOUR UNDERSTANDING OF SECTION 2.4 Four possible v_x - t graphs are shown for the two vehicles in Example 2.5. Which graph is correct?



ANSWER (b) The officer's x -acceleration is constant, so her v_x - t graph is a straight line. The motorist is moving faster than the car when the two vehicles meet at $t = 10$ s.

Figure 2.22 Multiflash photo of a freely falling ball.



2.5 FREELY FALLING OBJECTS

The most familiar example of motion with (nearly) constant acceleration is an object falling under the influence of the earth's gravitational attraction. Such motion has held the attention of philosophers and scientists since ancient times. In the fourth century B.C., Aristotle thought (erroneously) that heavy objects fall faster than light objects, in proportion to their weight. Nineteen centuries later, Galileo (see Section 1.1) argued that an object should fall with a downward acceleration that is constant and independent of its weight.

Experiment shows that if the effects of the air can be ignored, Galileo is right; all objects at a particular location fall with the same downward acceleration, regardless of their size or weight. If in addition the distance of the fall is small compared with the radius of the earth, and if we ignore small effects due to the earth's rotation, the acceleration is constant. The idealized motion that results under all of these assumptions is called **free fall**, although it includes rising as well as falling motion. (In Chapter 3 we'll extend the discussion of free fall to include the motion of projectiles, which move both vertically and horizontally.)

Figure 2.22 is a photograph of a falling ball made with a stroboscopic light source that produces a series of short, intense flashes at equal time intervals. As each flash occurs, an image of the ball at that instant is recorded on the photograph. The increasing spacing between successive images in Fig. 2.22 indicates that the ball is accelerating downward. Careful measurement shows that the velocity change is the same in each time interval, so the acceleration of the freely falling ball is constant.

The constant acceleration of a freely falling object is called the **acceleration due to gravity**, and we denote its magnitude with the letter g . We'll frequently use the approximate value of g at or near the earth's surface:

$$g = 9.80 \text{ m/s}^2 = 980 \text{ cm/s}^2 = 32.2 \text{ ft/s}^2 \quad (\text{approximate value near the earth's surface})$$

The exact value varies with location, so we'll often give the value of g at the earth's surface to only two significant figures as 9.8 m/s^2 . On the moon's surface, the acceleration due to gravity is caused by the attractive force of the moon rather than the earth, and $g = 1.6 \text{ m/s}^2$. Near the surface of the sun, $g = 270 \text{ m/s}^2$.

CAUTION g is always a positive number Because g is the *magnitude* of a vector quantity, it is always a *positive* number. If you take the positive y -direction to be upward, as we do in most situations involving free fall, the y -component of the acceleration is negative and equal to $-g$. Be careful with the sign of g , or you'll have trouble with free-fall problems. **|**

In the following examples we use the constant-acceleration equations developed in Section 2.4. Review Problem-Solving Strategy 2.1 in that section before you study the next examples.

EXAMPLE 2.6 A freely falling coin

A one-euro coin is dropped from the Leaning Tower of Pisa and falls freely from rest. What are its position and velocity after 1.0 s, 2.0 s, and 3.0 s? Ignore air resistance.

IDENTIFY and SET UP “Falls freely” means “falls with constant acceleration due to gravity,” so we can use the constant-acceleration equations. The right side of **Fig. 2.23** shows our motion diagram for the coin. The motion is vertical, so we use a vertical coordinate axis and call the coordinate y instead of x . We take the origin at the starting point and the *upward* direction as positive. Both the initial coordinate y_0 and initial y -velocity v_{0y} are zero. The y -acceleration is downward (in the negative y -direction), so $a_y = -g = -9.8 \text{ m/s}^2$. (Remember

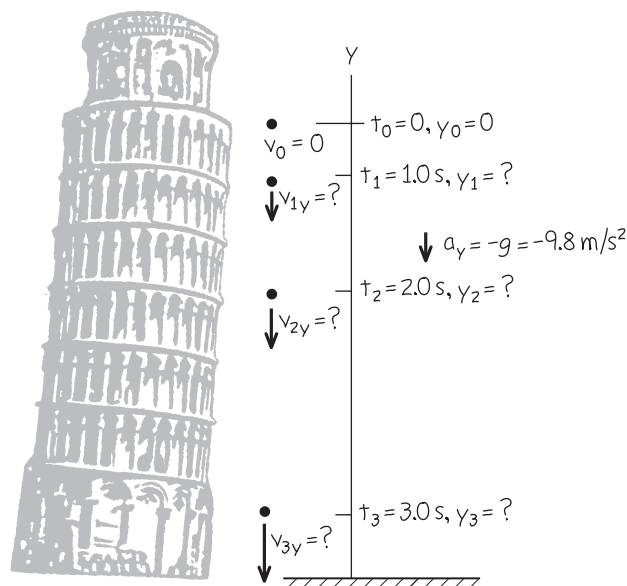
that g is a positive quantity.) Our target variables are the values of y and v_y at the three given times. To find these, we use Eqs. (2.12) and (2.8) with x replaced by y . Our choice of the upward direction as positive means that all positions and velocities we calculate will be negative.

EXECUTE At a time t after the coin is dropped, its position and y -velocity are

$$y = y_0 + v_{0y}t + \frac{1}{2}a_yt^2 = 0 + 0 + \frac{1}{2}(-g)t^2 = (-4.9 \text{ m/s}^2)t^2$$

$$v_y = v_{0y} + a_yt = 0 + (-g)t = (-9.8 \text{ m/s}^2)t$$

Figure 2.23 A coin freely falling from rest.



When $t = 1.0$ s, $y = (-4.9 \text{ m/s}^2)(1.0 \text{ s})^2 = -4.9 \text{ m}$ and $v_y = (-9.8 \text{ m/s}^2)(1.0 \text{ s}) = -9.8 \text{ m/s}$; after 1.0 s, the coin is 4.9 m below the origin (y is negative) and has a downward velocity (v_y is negative) with magnitude 9.8 m/s.

We can find the positions and y -velocities at 2.0 s and 3.0 s in the same way. The results are $y = -20 \text{ m}$ and $v_y = -20 \text{ m/s}$ at $t = 2.0$ s, and $y = -44 \text{ m}$ and $v_y = -29 \text{ m/s}$ at $t = 3.0$ s.

EVALUATE All our answers are negative, as we expected. If we had chosen the positive y -axis to point downward, the acceleration would have been $a_y = +g$ and all our answers would have been positive.

KEYCONCEPT By using one or more of the four equations in Table 2.5 with x replaced by y , the positive y -direction chosen to be upward, and acceleration $a_y = -g$, you can solve any free-fall problem.

EXAMPLE 2.7 Up-and-down motion in free fall

WITH VARIATION PROBLEMS

You throw a ball vertically upward from the roof of a tall building. The ball leaves your hand at a point even with the roof railing with an upward speed of 15.0 m/s; the ball is then in free fall. (We ignore air resistance.) On its way back down, it just misses the railing. Find (a) the ball's position and velocity 1.00 s and 4.00 s after leaving your hand; (b) the ball's velocity when it is 5.00 m above the railing; (c) the maximum height reached; (d) the ball's acceleration when it is at its maximum height.

IDENTIFY and SET UP The words “in free fall” mean that the acceleration is due to gravity, which is constant. Our target variables are position [in parts (a) and (c)], velocity [in parts (a) and (b)], and acceleration [in part (d)]. We take the origin at the point where the ball leaves your hand, and take the positive direction to be upward (**Fig. 2.24**). The initial position y_0 is zero, the initial y -velocity v_{0y} is +15.0 m/s, and the y -acceleration is $a_y = -g = -9.80 \text{ m/s}^2$. In part (a), as in Example 2.6, we'll use Eqs. (2.12) and (2.8) to find the position and velocity as functions of time. In part (b) we must find the velocity at a given *position* (no time is given), so we'll use Eq. (2.13).

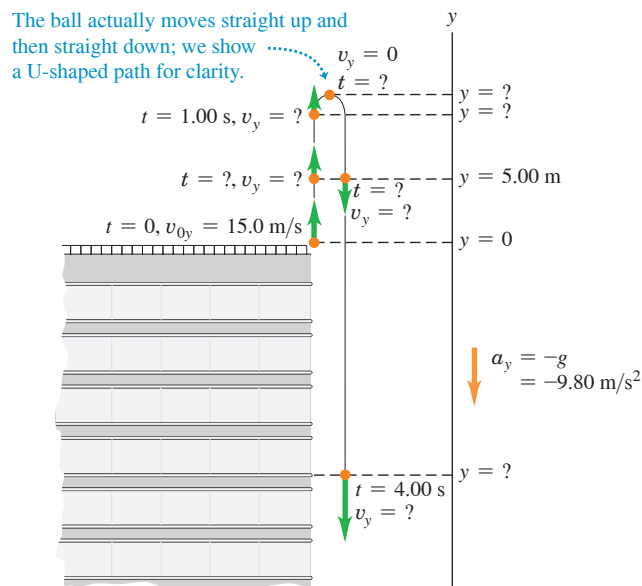
Figure 2.25 (next page) shows the y - t and v_y - t graphs for the ball. The y - t graph is a concave-down parabola that rises and then falls, and the v_y - t graph is a downward-sloping straight line. Note that the ball's velocity is zero when it is at its highest point.

EXECUTE (a) The position and y -velocity at time t are given by Eqs. (2.12) and (2.8) with x 's replaced by y 's:

$$\begin{aligned} y &= y_0 + v_{0y}t + \frac{1}{2}a_yt^2 = y_0 + v_{0y}t + \frac{1}{2}(-g)t^2 \\ &= (0) + (15.0 \text{ m/s})t + \frac{1}{2}(-9.80 \text{ m/s}^2)t^2 \\ v_y &= v_{0y} + a_yt = v_{0y} + (-g)t \\ &= 15.0 \text{ m/s} + (-9.80 \text{ m/s}^2)t \end{aligned}$$

When $t = 1.00$ s, these equations give $y = +10.1 \text{ m}$ and $v_y = +5.2 \text{ m/s}$. That is, the ball is 10.1 m above the origin (y is positive) and

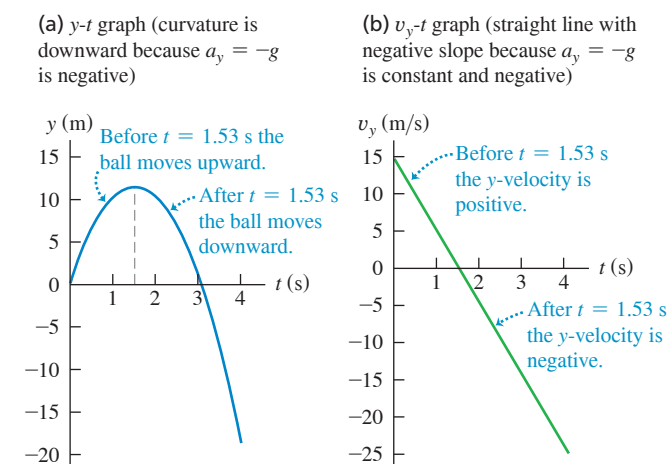
Figure 2.24 Position and velocity of a ball thrown vertically upward.



moving upward (v_y is positive) with a speed of 5.2 m/s. This is less than the initial speed because the ball slows as it ascends. When $t = 4.00$ s, those equations give $y = -18.4 \text{ m}$ and $v_y = -24.2 \text{ m/s}$. The ball has passed its highest point and is 18.4 m below the origin (y is negative). It is moving downward (v_y is negative) with a speed of 24.2 m/s.

Continued

Figure 2.25 (a) Position and (b) velocity as functions of time for a ball thrown upward with an initial speed of 15.0 m/s.



(b) The y -velocity at any position y is given by Eq. (2.13) with x 's replaced by y 's:

$$\begin{aligned} v_y^2 &= v_{0y}^2 + 2a_y(y - y_0) = v_{0y}^2 + 2(-g)(y - 0) \\ &= (15.0 \text{ m/s})^2 + 2(-9.80 \text{ m/s}^2)y \end{aligned}$$

When the ball is 5.00 m above the origin we have $y = +5.00$ m, so

$$\begin{aligned} v_y^2 &= (15.0 \text{ m/s})^2 + 2(-9.80 \text{ m/s}^2)(5.00 \text{ m}) = 127 \text{ m}^2/\text{s}^2 \\ v_y &= \pm 11.3 \text{ m/s} \end{aligned}$$

We get *two* values of v_y because the ball passes through the point $y = +5.00$ m twice, once on the way up (so v_y is positive) and once on the way down (so v_y is negative) (see Fig. 2.24). Note that the speed of the ball is 11.3 m/s each time it passes through this point.

(c) At the instant at which the ball reaches its maximum height y_1 , its y -velocity is momentarily zero: $v_y = 0$. We use Eq. (2.13) to find y_1 . With $v_y = 0$, $y_0 = 0$, and $a_y = -g$, we get

$$\begin{aligned} 0 &= v_{0y}^2 + 2(-g)(y_1 - 0) \\ y_1 &= \frac{v_{0y}^2}{2g} = \frac{(15.0 \text{ m/s})^2}{2(9.80 \text{ m/s}^2)} = +11.5 \text{ m} \end{aligned}$$

(d) **CAUTION A free-fall misconception** It's a common misconception that at the highest point of free-fall motion, where the velocity is zero, the acceleration is also zero. If this were so, once the ball reached the highest point it would hang there suspended in midair! Remember that acceleration is the rate of change of velocity, and the ball's velocity is continuously changing. At every point, including the highest point, and at any velocity, including zero, the acceleration in free fall is always $a_y = -g = -9.80 \text{ m/s}^2$.

EVALUATE A useful way to check any free-fall problem is to draw the y - t and v_y - t graphs, as we did in Fig. 2.25. Note that these are graphs of Eqs. (2.12) and (2.8), respectively. Given the initial position, initial velocity, and acceleration, you can easily create these graphs by using a graphing calculator app or an online math program.

KEYCONCEPT If a freely falling object passes a given point at two different times, once moving upward and once moving downward, its speed will be the same at both times.

EXAMPLE 2.8 Two solutions or one?

WITH VARIATION PROBLEMS

At what time after being released has the ball in Example 2.7 fallen 5.00 m below the roof railing?

IDENTIFY and SET UP We treat this as in Example 2.7, so y_0 , v_{0y} , and $a_y = -g$ have the same values as there. Now, however, the target variable is the time at which the ball is at $y = -5.00$ m. The best equation to use is Eq. (2.12) with a_y replaced by $-g$, which gives the position y as a function of time t :

$$y = y_0 + v_{0y}t + \frac{1}{2}(-g)t^2$$

This is a *quadratic* equation for t , which we want to solve for the value of t when $y = -5.00$ m.

EXECUTE We rearrange the equation so that it has the standard form of a quadratic equation for an unknown x , $Ax^2 + Bx + C = 0$:

$$\left(\frac{1}{2}g\right)t^2 + (-v_{0y})t + (y - y_0) = At^2 + Bt + C = 0$$

By comparison, we identify $A = \frac{1}{2}g$, $B = -v_{0y}$, and $C = y - y_0$. The quadratic formula (see Appendix B) tells us that this equation has *two* solutions:

$$\begin{aligned} t &= \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \\ &= \frac{-(-v_{0y}) \pm \sqrt{(-v_{0y})^2 - 4\left(\frac{1}{2}g\right)(y - y_0)}}{2\left(\frac{1}{2}g\right)} \\ &= \frac{v_{0y} \pm \sqrt{v_{0y}^2 - 2g(y - y_0)}}{g} \end{aligned}$$

Substituting the values $y_0 = 0$, $v_{0y} = +15.0 \text{ m/s}$, $g = 9.80 \text{ m/s}^2$, and $y = -5.00$ m, we find

$$t = \frac{(15.0 \text{ m/s}) \pm \sqrt{(15.0 \text{ m/s})^2 - 2(9.80 \text{ m/s}^2)(-5.00 \text{ m} - 0)}}{9.80 \text{ m/s}^2}$$

You can confirm that the numerical answers are $t = +3.36 \text{ s}$ and $t = -0.30 \text{ s}$. The answer $t = -0.30 \text{ s}$ doesn't make physical sense, since it refers to a time *before* the ball left your hand at $t = 0$. So the correct answer is $t = +3.36 \text{ s}$.

EVALUATE Why did we get a second, fictitious solution? The explanation is that constant-acceleration equations like Eq. (2.12) are based on the assumption that the acceleration is constant for *all* values of time, whether positive, negative, or zero. Hence the solution $t = -0.30 \text{ s}$ refers to an imaginary moment when a freely falling ball was 5.00 m below the roof railing and rising to meet your hand. Since the ball didn't leave your hand and go into free fall until $t = 0$, this result is pure fiction.

Repeat these calculations to find the times when the ball is 5.00 m *above* the origin ($y = +5.00 \text{ m}$). The two answers are $t = +0.38 \text{ s}$ and $t = +2.68 \text{ s}$. Both are positive values of t , and both refer to the real motion of the ball after leaving your hand. At the earlier time the ball passes through $y = +5.00 \text{ m}$ moving upward; at the later time it passes through this point moving downward. [Compare this with part (b) of Example 2.7, and again refer to Fig. 2.25a.]

You should also solve for the times when $y = +15.0 \text{ m}$. In this case, both solutions involve the square root of a negative number, so there are *no* real solutions. Again Fig. 2.25a shows why; we found in part

(c) of Example 2.7 that the ball's maximum height is $y = +11.5$ m, so it *never* reaches $y = +15.0$ m. While a quadratic equation such as Eq. (2.12) always has two solutions, in some situations one or both of the solutions aren't physically reasonable.

In this example we encountered a quadratic equation for the case of free fall. But Eq. (2.12) is a quadratic equation that applies to *all* cases of straight-line motion with constant acceleration, so you'll

need to exercise the same care in solving many problems of this kind.

KEYCONCEPT When the acceleration of an object is constant, its position as a function of time is given by a quadratic equation. Inspect the solutions to this equation to determine what they tell you about the problem you're solving.

TEST YOUR UNDERSTANDING OF SECTION 2.5 If you toss a ball upward with a certain initial speed, it falls freely and reaches a maximum height h a time t after it leaves your hand. (a) If you throw the ball upward with double the initial speed, what new maximum height does the ball reach? (i) $h\sqrt{2}$; (ii) $2h$; (iii) $4h$; (iv) $8h$; (v) $16h$. (b) If you throw the ball upward with double the initial speed, how long does it take to reach its new maximum height? (i) $t/2$; (ii) $t/\sqrt{2}$; (iii) t ; (iv) $t\sqrt{2}$; (v) $2t$.

ANSWER (a) (iii) Use Eq. (2.13) with x replaced by y and $a_y = -g$: $v_y^2 = v_{0y}^2 - 2g(y - y_0)$. The starting height is $y_0 = 0$ and the y -velocity at the maximum height $y = h$ is $v_y = 0$, so $0 = v_{0y}^2 - 2gh$ and $h = v_{0y}^2/2g$. If the initial y -velocity is increased by a factor of 2, the maximum height increases by a factor of $2^2 = 4$ and the ball goes to height $4h$. (b) (v) Use Eq. (2.8) with x replaced by y and $a_y = -g$: $v_y = v_{0y} - gt$. The y -velocity at the maximum height is $v_y = 0$, so $0 = v_{0y} - gt$ and $t = v_{0y}/g$. If the initial y -velocity is increased by a factor of 2, the time to reach the maximum height increases by a factor of 2 and becomes $2t$.

2.6 VELOCITY AND POSITION BY INTEGRATION

This section is intended for students who have already learned a little integral calculus. In Section 2.4 we analyzed the special case of straight-line motion with constant acceleration. When a_x is not constant, as is frequently the case, the equations that we derived in that section are no longer valid (**Fig. 2.26**). But even when a_x varies with time, we can still use the relationship $v_x = dx/dt$ to find the x -velocity v_x as a function of time if the position x is a known function of time. And we can still use $a_x = dv_x/dt$ to find the x -acceleration a_x as a function of time if the x -velocity v_x is a known function of time.

In many situations, however, position and velocity are not known functions of time, while acceleration is (**Fig. 2.27**). How can we find the position and velocity in straight-line motion from the acceleration function $a_x(t)$?

Figure 2.26 When you push a car's accelerator pedal to the floorboard, the resulting acceleration is *not* constant: The greater the car's speed, the more slowly it gains additional speed. A typical car takes twice as long to accelerate from 50 km/h to 100 km/h as it does to accelerate from 0 to 50 km/h.



Figure 2.27 The inertial navigation system (INS) on board a long-range airliner keeps track of the airliner's acceleration. Given the airliner's initial position and velocity before takeoff, the INS uses the acceleration data to calculate the airliner's position and velocity throughout the flight.



Figure 2.28 An a_x - t graph for a particle whose x -acceleration is not constant.

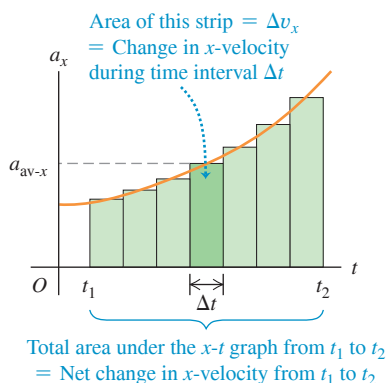


Figure 2.28 is a graph of x -acceleration versus time for a particle whose acceleration is not constant. We can divide the time interval between times t_1 and t_2 into many smaller subintervals, calling a typical one Δt . Let the average x -acceleration during Δt be $a_{\text{av-}x}$. From Eq. (2.4) the change in x -velocity Δv_x during Δt is

$$\Delta v_x = a_{\text{av-}x} \Delta t$$

Graphically, Δv_x equals the area of the shaded strip with height $a_{\text{av-}x}$ and width Δt —that is, the area under the curve between the left and right sides of Δt . The total change in x -velocity from t_1 to t_2 is the sum of the x -velocity changes Δv_x in the small subintervals. So the total x -velocity change is represented graphically by the *total* area under the a_x - t curve between the vertical lines t_1 and t_2 . (In Section 2.4 we showed this for the special case in which a_x is constant.)

In the limit that all the Δt 's become very small and they become very large in number, the value of $a_{\text{av-}x}$ for the interval from any time t to $t + \Delta t$ approaches the instantaneous x -acceleration a_x at time t . In this limit, the area under the a_x - t curve is the *integral* of a_x (which is in general a function of t) from t_1 to t_2 . If v_{1x} is the x -velocity of the particle at time t_1 and v_{2x} is the velocity at time t_2 , then

$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x dt \quad (2.15)$$

The change in the x -velocity v_x is the time integral of the x -acceleration a_x .

We can carry out exactly the same procedure with the curve of x -velocity versus time. If x_1 is a particle's position at time t_1 and x_2 is its position at time t_2 , from Eq. (2.2) the displacement Δx during a small time interval Δt is equal to $v_{\text{av-}x} \Delta t$, where $v_{\text{av-}x}$ is the average x -velocity during Δt . The total displacement $x_2 - x_1$ during the interval $t_2 - t_1$ is given by

$$x_2 - x_1 = \int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v_x dt \quad (2.16)$$

The change in position x —that is, the displacement—is the time integral of x -velocity v_x . Graphically, the displacement between times t_1 and t_2 is the area under the v_x - t curve between those two times. [This is the same result that we obtained in Section 2.4 for the special case in which v_x is given by Eq. (2.8).]

If $t_1 = 0$ and t_2 is any later time t , and if x_0 and v_{0x} are the position and velocity, respectively, at time $t = 0$, then we can rewrite Eqs. (2.15) and (2.16) as

$$v_x = v_{0x} + \underbrace{\int_0^t a_x dt}_{\text{Integral of the } x\text{-acceleration of the particle from time 0 to time } t} \quad (2.17)$$

x -velocity of a particle at time t x -velocity of the particle at time 0

$$x = x_0 + \underbrace{\int_0^t v_x dt}_{\text{Integral of the } x\text{-velocity of the particle from time 0 to time } t} \quad (2.18)$$

Position of a particle at time t Position of the particle at time 0

If we know the x -acceleration a_x as a function of time and we know the initial velocity v_{0x} , we can use Eq. (2.17) to find the x -velocity v_x at any time; that is, we can find v_x as a function of time. Once we know this function, and given the initial position x_0 , we can use Eq. (2.18) to find the position x at any time.

EXAMPLE 2.9 Motion with changing acceleration

Sally is driving along a straight highway. At $t = 0$, when she is moving at 10 m/s in the positive x -direction, she passes a signpost at $x = 50$ m. Her x -acceleration as a function of time is

$$a_x = 2.0 \text{ m/s}^2 - (0.10 \text{ m/s}^3)t$$

(a) Find her x -velocity v_x and position x as functions of time. (b) When is her x -velocity greatest? (c) What is that maximum x -velocity? (d) Where is the car when it reaches that maximum x -velocity?

IDENTIFY and SET UP The x -acceleration is a function of time, so we *cannot* use the constant-acceleration formulas of Section 2.4. Instead, we use Eq. (2.17) to obtain an expression for v_x as a function of time, and then use that result in Eq. (2.18) to find an expression for x as a function of t . We'll then be able to answer a variety of questions about the motion.

EXECUTE (a) At $t = 0$, Sally's position is $x_0 = 50$ m and her x -velocity is $v_{0x} = 10$ m/s. To use Eq. (2.17), we note that the integral of t^n (except for $n = -1$) is $\int t^n dt = \frac{1}{n+1}t^{n+1}$. Hence

$$\begin{aligned} v_x &= 10 \text{ m/s} + \int_0^t [2.0 \text{ m/s}^2 - (0.10 \text{ m/s}^3)t] dt \\ &= 10 \text{ m/s} + (2.0 \text{ m/s}^2)t - \frac{1}{2}(0.10 \text{ m/s}^3)t^2 \end{aligned}$$

Now we use Eq. (2.18) to find x as a function of t :

$$\begin{aligned} x &= 50 \text{ m} + \int_0^t [10 \text{ m/s} + (2.0 \text{ m/s}^2)t - \frac{1}{2}(0.10 \text{ m/s}^3)t^2] dt \\ &= 50 \text{ m} + (10 \text{ m/s})t + \frac{1}{2}(2.0 \text{ m/s}^2)t^2 - \frac{1}{6}(0.10 \text{ m/s}^3)t^3 \end{aligned}$$

Figure 2.29 shows graphs of a_x , v_x , and x as functions of time as given by the previous equations. Note that for any time t , the slope of the v_x - t graph equals the value of a_x and the slope of the x - t graph equals the value of v_x .

(b) The maximum value of v_x occurs when the x -velocity stops increasing and begins to decrease. At that instant, $dv_x/dt = a_x = 0$. So we set the expression for a_x equal to zero and solve for t :

$$\begin{aligned} 0 &= 2.0 \text{ m/s}^2 - (0.10 \text{ m/s}^3)t \\ t &= \frac{2.0 \text{ m/s}^2}{0.10 \text{ m/s}^3} = 20 \text{ s} \end{aligned}$$

(c) We find the maximum x -velocity by substituting $t = 20$ s, the time from part (b) when velocity is maximum, into the equation for v_x from part (a):

$$\begin{aligned} v_{\text{max-}x} &= 10 \text{ m/s} + (2.0 \text{ m/s}^2)(20 \text{ s}) - \frac{1}{2}(0.10 \text{ m/s}^3)(20 \text{ s})^2 \\ &= 30 \text{ m/s} \end{aligned}$$

(d) To find the car's position at the time that we found in part (b), we substitute $t = 20$ s into the expression for x from part (a):

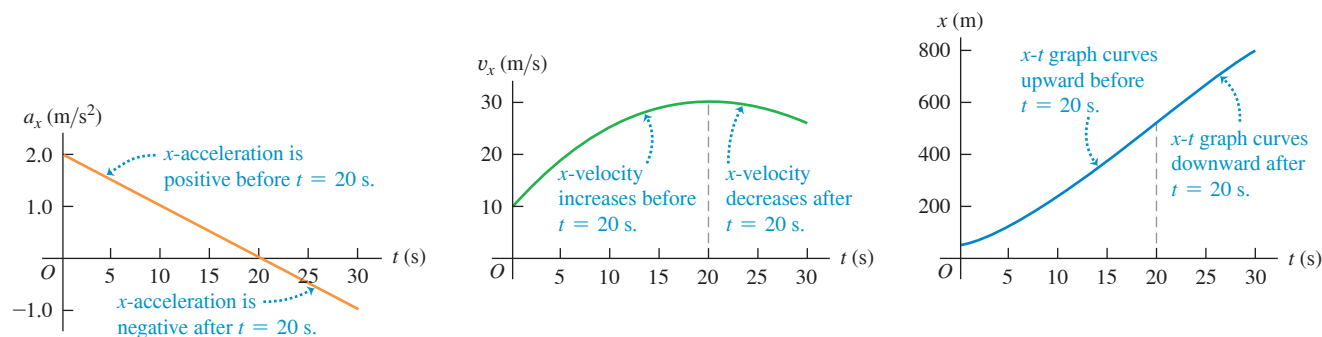
$$\begin{aligned} x &= 50 \text{ m} + (10 \text{ m/s})(20 \text{ s}) + \frac{1}{2}(2.0 \text{ m/s}^2)(20 \text{ s})^2 - \frac{1}{6}(0.10 \text{ m/s}^3)(20 \text{ s})^3 \\ &= 517 \text{ m} \end{aligned}$$

EVALUATE Figure 2.29 helps us interpret our results. The left-hand graph shows that a_x is positive between $t = 0$ and $t = 20$ s and negative after that. It is zero at $t = 20$ s, the time at which v_x is maximum (the high point in the middle graph). The car speeds up until $t = 20$ s (because v_x and a_x have the same sign) and slows down after $t = 20$ s (because v_x and a_x have opposite signs).

Since v_x is maximum at $t = 20$ s, the x - t graph (the right-hand graph in Fig. 2.29) has its maximum positive slope at this time. Note that the x - t graph is concave up (curved upward) from $t = 0$ to $t = 20$ s, when a_x is positive. The graph is concave down (curved downward) after $t = 20$ s, when a_x is negative.

KEYCONCEPT If the acceleration in straight-line motion is not constant but is a known function of time, you can find the velocity and position as functions of time by integration.

Figure 2.29 The position, velocity, and acceleration of the car in Example 2.9 as functions of time. Can you show that if this motion continues, the car will stop at $t = 44.5$ s?



TEST YOUR UNDERSTANDING OF SECTION 2.6 If the x -acceleration a_x of an object moving in straight-line motion is increasing with time, will the v_x - t graph be (i) a straight line, (ii) concave up (i.e., with an upward curvature), or (iii) concave down (i.e., with a downward curvature)?

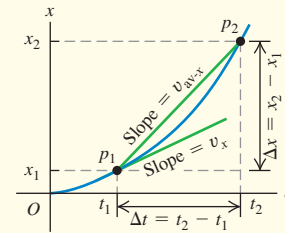
ANSWER (ii) The acceleration a_x is equal to the slope of the v_x - t graph. If a_x is increasing, the slope of the v_x - t graph is also increasing and the graph is concave up.

CHAPTER 2 SUMMARY

Straight-line motion, average and instantaneous x -velocity: When a particle moves along a straight line, we describe its position with respect to an origin O by means of a coordinate such as x . The particle's average x -velocity $v_{\text{av-}x}$ during a time interval $\Delta t = t_2 - t_1$ is equal to its displacement $\Delta x = x_2 - x_1$ divided by Δt . The instantaneous x -velocity v_x at any time t is equal to the average x -velocity over the time interval from t to $t + \Delta t$ in the limit that Δt goes to zero. Equivalently, v_x is the derivative of the position function with respect to time. (See Example 2.1.)

$$v_{\text{av-}x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \quad (2.2)$$

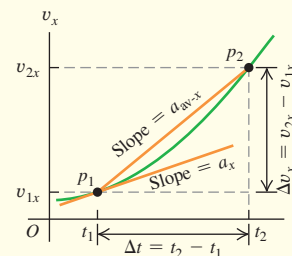
$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (2.3)$$



Average and instantaneous x -acceleration: The average x -acceleration $a_{\text{av-}x}$ during a time interval Δt is equal to the change in velocity $\Delta v_x = v_{2x} - v_{1x}$ during that time interval divided by Δt . The instantaneous x -acceleration a_x is the limit of $a_{\text{av-}x}$ as Δt goes to zero, or the derivative of v_x with respect to t . (See Examples 2.2 and 2.3.)

$$a_{\text{av-}x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} \quad (2.4)$$

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} \quad (2.5)$$



Straight-line motion with constant acceleration: When the x -acceleration is constant, four equations relate the position x and the x -velocity v_x at any time t to the initial position x_0 , the initial x -velocity v_{0x} (both measured at time $t = 0$), and the x -acceleration a_x . (See Examples 2.4 and 2.5.)

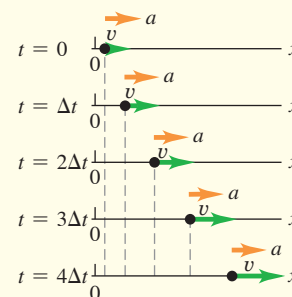
Constant x -acceleration only:

$$v_x = v_{0x} + a_x t \quad (2.8)$$

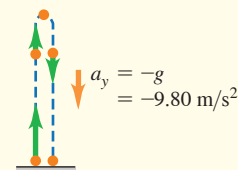
$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2 \quad (2.12)$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0) \quad (2.13)$$

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t \quad (2.14)$$



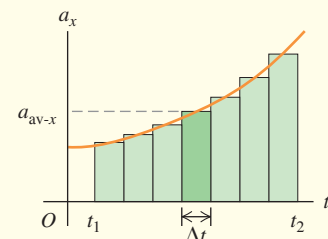
Freely falling objects: Free fall (vertical motion without air resistance, so only gravity affects the motion) is a case of motion with constant acceleration. The magnitude of the acceleration due to gravity is a positive quantity, g . The acceleration of an object in free fall is always downward. (See Examples 2.6–2.8.)

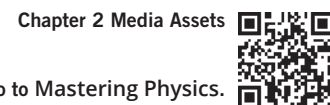


Straight-line motion with varying acceleration: When the acceleration is not constant but is a known function of time, we can find the velocity and position as functions of time by integrating the acceleration function. (See Example 2.9.)

$$v_x = v_{0x} + \int_0^t a_x dt \quad (2.17)$$

$$x = x_0 + \int_0^t v_x dt \quad (2.18)$$





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 2.5** (Section 2.4) before attempting these problems.

VP2.5.1 A sports car starts from rest at an intersection and accelerates toward the east on a straight road at 8.0 m/s^2 . Just as the sports car starts to move, a bus traveling east at a constant 18 m/s on the same straight road passes the sports car. When the sports car catches up with and passes the bus, (a) how much time has elapsed and (b) how far has the sports car traveled?

VP2.5.2 A car is traveling on a straight road at a constant 30.0 m/s , which is faster than the speed limit. Just as the car passes a police motorcycle that is stopped at the side of the road, the motorcycle accelerates forward in pursuit. The motorcycle passes the car 12.5 s after starting from rest. (a) What is the acceleration of the motorcycle (assumed to be constant)? (b) How far does the motorcycle travel before it passes the car?

VP2.5.3 A police car is traveling north on a straight road at a constant 20.0 m/s . An SUV traveling north at 30.0 m/s passes the police car. The driver of the SUV suspects he may be exceeding the speed limit, so just as he passes the police car he lets the SUV slow down at a constant 1.80 m/s^2 . (a) How much time elapses from when the SUV passes the police car to when the police car passes the SUV? (b) What distance does the SUV travel during this time? (c) What is the speed of the SUV when the police car passes it?

VP2.5.4 At $t = 0$ a truck starts from rest at $x = 0$ and speeds up in the positive x -direction on a straight road with acceleration a_T . At the same time, $t = 0$, a car is at $x = 0$ and traveling in the positive x -direction with speed v_C . The car has a constant negative x -acceleration: $a_{\text{car-}x} = -a_C$, where a_C is a positive quantity. (a) At what time does the truck pass the car? (b) At what x -coordinate does the truck pass the car?

Be sure to review **EXAMPLE 2.7** (Section 2.5) before attempting these problems.

VP2.7.1 You throw a ball straight up from the edge of a cliff. It leaves your hand moving at 12.0 m/s . Air resistance can be neglected. Take the positive y -direction to be upward, and choose $y = 0$ to be the point where the ball leaves your hand. Find the ball's position and velocity (a) 0.300 s after it leaves your hand and (b) 2.60 s after it leaves your hand. At each time state whether the ball is above or below your hand and whether it is moving upward or downward.

VP2.7.2 You throw a stone straight down from the top of a tall tower. It leaves your hand moving at 8.00 m/s . Air resistance can be neglected. Take the positive y -direction to be upward, and choose $y = 0$ to be the

point where the stone leaves your hand. (a) Find the stone's position and velocity 1.50 s after it leaves your hand. (b) Find the stone's velocity when it is 8.00 m below your hand.

VP2.7.3 You throw a football straight up. Air resistance can be neglected. (a) When the football is 4.00 m above where it left your hand, it is moving upward at 0.500 m/s . What was the speed of the football when it left your hand? (b) How much time elapses from when the football leaves your hand until it is 4.00 m above your hand?

VP2.7.4 You throw a tennis ball straight up. Air resistance can be neglected. (a) The maximum height above your hand that the ball reaches is H . At what speed does the ball leave your hand? (b) What is the speed of the ball when it is a height $H/2$ above your hand? Express your answer as a fraction of the speed at which it left your hand. (c) At what height above your hand is the speed of the ball half as great as when it left your hand? Express your answer in terms of H .

Be sure to review **EXAMPLE 2.8** (Section 2.5) before attempting these problems.

VP2.8.1 You throw a rock straight up from the edge of a cliff. It leaves your hand at time $t = 0$ moving at 12.0 m/s . Air resistance can be neglected. (a) Find both times at which the rock is 4.00 m above where it left your hand. (b) Find the time when the rock is 4.00 m below where it left your hand.

VP2.8.2 You throw a basketball straight down from the roof of a building. The basketball leaves your hand at time $t = 0$ moving at 9.00 m/s . Air resistance can be neglected. Find the time when the ball is 5.00 m below where it left your hand.

VP2.8.3 You throw an apple straight up. The apple leaves your hand at time $t = 0$ moving at 5.50 m/s . Air resistance can be neglected. (a) How many times (two, one, or none) does the apple pass through a point 1.30 m above your hand? If the apple does pass through this point, at what times t does it do so, and is the apple moving upward or downward at each of these times? (b) How many times (two, one, or none) does the apple pass through a point 1.80 m above your hand? If the apple does pass through this point, at what times t does it do so, and is the apple moving upward or downward at each of these times?

VP2.8.4 You throw an orange straight up. The orange leaves your hand at time $t = 0$ moving at speed v_0 . Air resistance can be neglected. (a) At what time(s) is the orange at a height $v_0^2/2g$ above the point where it left your hand? At these time(s) is the orange moving upward, downward, or neither? (b) At what time(s) is the orange at a height $3v_0^2/8g$ above the point where it left your hand? At these time(s) is the orange moving upward, downward, or neither?

BRIDGING PROBLEM The Fall of a Superhero

The superhero Green Lantern steps from the top of a tall building. He falls freely from rest to the ground, falling half the total distance to the ground during the last 1.00 s of his fall (**Fig. 2.30**). What is the height h of the building?

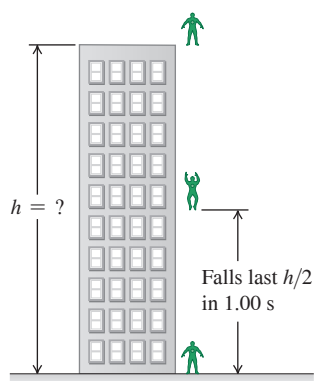
SOLUTION GUIDE

IDENTIFY and SET UP

1. You're told that Green Lantern falls freely from rest. What does this imply about his acceleration? About his initial velocity?

2. Choose the direction of the positive y -axis. It's easiest to make the same choice we used for freely falling objects in Section 2.5.
3. You can divide Green Lantern's fall into two parts: from the top of the building to the halfway point and from the halfway point to the ground. You know that the second part of the fall lasts 1.00 s . Decide what you would need to know about Green Lantern's motion at the halfway point in order to solve for the target variable h . Then choose two equations, one for the first part of the fall and one for the second part, that you'll use

Figure 2.30 Green Lantern in free fall.



together to find an expression for h . (There are several pairs of equations that you could choose.)

EXECUTE

- Use your two equations to solve for the height h . Heights are always positive numbers, so your answer should be positive.

EVALUATE

- To check your answer for h , use one of the free-fall equations to find how long it takes Green Lantern to fall (i) from the top of the building to half the height and (ii) from the top of the building to the ground. If your answer for h is correct, time (ii) should be 1.00 s greater than time (i). If it isn't, go back and look for errors in how you found h .

PROBLEMS

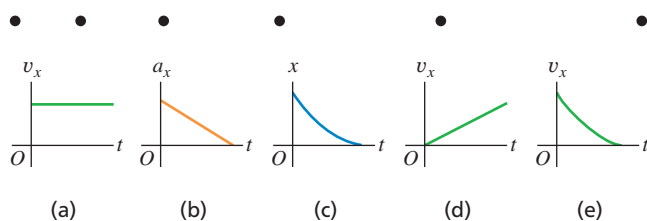
•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q2.1 Does the speedometer of a car measure speed or velocity? Explain.

Q2.2 The black dots at the top of **Fig. Q2.2** represent a series of high-speed photographs of an insect flying in a straight line from left to right (in the positive x -direction). Which of the graphs in **Fig. Q2.2** most plausibly depicts this insect's motion?

Figure Q2.2



Q2.3 Can an object with constant acceleration reverse its direction of travel? Can it reverse its direction *twice*? In both cases, explain your reasoning.

Q2.4 Under what conditions is average velocity equal to instantaneous velocity?

Q2.5 Is it possible for an object to be (a) slowing down while its acceleration is increasing in magnitude; (b) speeding up while its acceleration is decreasing? In both cases, explain your reasoning.

Q2.6 Under what conditions does the magnitude of the average velocity equal the average speed?

Q2.7 When a Dodge Viper is at Elwood's Car Wash, a BMW Z3 is at Elm and Main. Later, when the Dodge reaches Elm and Main, the BMW reaches Elwood's Car Wash. How are the cars' average velocities between these two times related?

Q2.8 A driver in Massachusetts was sent to traffic court for speeding. The evidence against the driver was that a policewoman observed the driver's car alongside a second car at a certain moment, and the policewoman had already clocked the second car going faster than the speed limit. The driver argued, "The second car was passing me. I was not

speeding." The judge ruled against the driver because, in the judge's words, "If two cars were side by side, both of you were speeding." If you were a lawyer representing the accused driver, how would you argue this case?

Q2.9 Can you have zero displacement and nonzero average velocity? Zero displacement and nonzero velocity? Illustrate your answers on an x - t graph.

Q2.10 Can you have zero acceleration and nonzero velocity? Use a v_x - t graph to explain.

Q2.11 Can you have zero velocity and nonzero average acceleration? Zero velocity and nonzero acceleration? Use a v_x - t graph to explain, and give an example of such motion.

Q2.12 An automobile is traveling west. Can it have a velocity toward the west and at the same time have an acceleration toward the east? Under what circumstances?

Q2.13 The official's truck in **Fig. 2.2** is at $x_1 = 277$ m at $t_1 = 16.0$ s and is at $x_2 = 19$ m at $t_2 = 25.0$ s. (a) Sketch *two* different possible x - t graphs for the motion of the truck. (b) Does the average velocity v_{av-x} during the time interval from t_1 to t_2 have the same value for both of your graphs? Why or why not?

Q2.14 Under constant acceleration the average velocity of a particle is half the sum of its initial and final velocities. Is this still true if the acceleration is *not* constant? Explain.

Q2.15 You throw a baseball straight up in the air so that it rises to a maximum height much greater than your height. Is the magnitude of the ball's acceleration greater while it is being thrown or after it leaves your hand? Explain.

Q2.16 Prove these statements: (a) As long as you can ignore the effects of the air, if you throw anything vertically upward, it will have the same speed when it returns to the release point as when it was released. (b) The time of flight will be twice the time it takes to get to its highest point.

Q2.17 A dripping water faucet steadily releases drops 1.0 s apart. As these drops fall, does the distance between them increase, decrease, or remain the same? Prove your answer.

Q2.18 If you know the initial position and initial velocity of a vehicle and have a record of the acceleration at each instant, can you compute the vehicle's position after a certain time? If so, explain how this might be done.

Q2.19 From the top of a tall building, you throw one ball straight up with speed v_0 and one ball straight down with speed v_0 . (a) Which ball has the greater speed when it reaches the ground? (b) Which ball gets to the ground first? (c) Which ball has a greater displacement when it reaches the ground? (d) Which ball has traveled the greater distance when it hits the ground?

Q2.20 You run due east at a constant speed of 3.00 m/s for a distance of 120.0 m and then continue running east at a constant speed of 5.00 m/s for another 120.0 m. For the total 240.0 m run, is your average velocity 4.00 m/s, greater than 4.00 m/s, or less than 4.00 m/s? Explain.

Q2.21 An object is thrown straight up into the air and feels no air resistance. How can the object have an acceleration when it has stopped moving at its highest point?

Q2.22 When you drop an object from a certain height, it takes time T to reach the ground with no air resistance. If you dropped it from three times that height, how long (in terms of T) would it take to reach the ground?

EXERCISES

Section 2.1 Displacement, Time, and Average Velocity

2.1 • A car travels in the $+x$ -direction on a straight and level road. For the first 4.00 s of its motion, the average velocity of the car is $v_{av-x} = 6.25$ m/s. How far does the car travel in 4.00 s?

2.2 •• In an experiment, a shearwater (a seabird) was taken from its nest, flown 5150 km away, and released. The bird found its way back to its nest 13.5 days after release. If we place the origin at the nest and extend the $+x$ -axis to the release point, what was the bird's average velocity in m/s (a) for the return flight and (b) for the whole episode, from leaving the nest to returning?

2.3 •• Trip Home. You normally drive on the freeway between San Diego and Los Angeles at an average speed of 105 km/h (65 mi/h), and the trip takes 1 h and 50 min. On a Friday afternoon, however, heavy traffic slows you down and you drive the same distance at an average speed of only 70 km/h (43 mi/h). How much longer does the trip take?

2.4 •• From Pillar to Post. Starting from a pillar, you run 200 m east (the $+x$ -direction) at an average speed of 5.0 m/s and then run 280 m west at an average speed of 4.0 m/s to a post. Calculate (a) your average speed from pillar to post and (b) your average velocity from pillar to post.

2.5 • Starting from the front door of a ranch house, you walk 60.0 m due east to a windmill, turn around, and then slowly walk 40.0 m west to a bench, where you sit and watch the sunrise. It takes you 28.0 s to walk from the house to the windmill and then 36.0 s to walk from the windmill to the bench. For the entire trip from the front door to the bench, what are your (a) average velocity and (b) average speed?

2.6 •• A Honda Civic travels in a straight line along a road. The car's distance x from a stop sign is given as a function of time t by the equation $x(t) = \alpha t^2 - \beta t^3$, where $\alpha = 1.50$ m/s² and $\beta = 0.0500$ m/s³. Calculate the average velocity of the car for each time interval: (a) $t = 0$ to $t = 2.00$ s; (b) $t = 0$ to $t = 4.00$ s; (c) $t = 2.00$ s to $t = 4.00$ s.

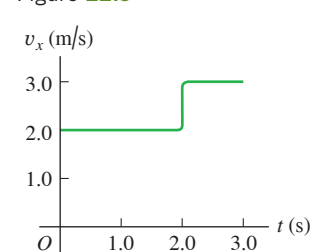
Section 2.2 Instantaneous Velocity

2.7 • CALC A car is stopped at a traffic light. It then travels along a straight road such that its distance from the light is given by $x(t) = bt^2 - ct^3$, where $b = 2.40$ m/s² and $c = 0.120$ m/s³. (a) Calculate the average velocity of the car for the time interval $t = 0$ to $t = 10.0$ s. (b) Calculate the instantaneous velocity of the car at $t = 0$, $t = 5.0$ s, and $t = 10.0$ s. (c) How long after starting from rest is the car again at rest?

2.8 • CALC A bird is flying due east. Its distance from a tall building is given by $x(t) = 28.0 \text{ m} + (12.4 \text{ m/s})t - (0.0450 \text{ m/s}^3)t^3$. What is the instantaneous velocity of the bird when $t = 8.00$ s?

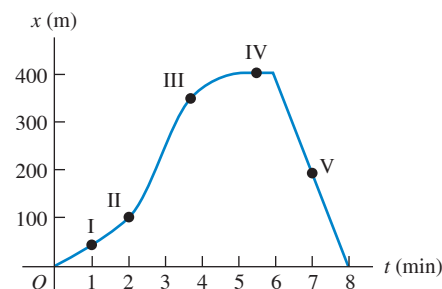
2.9 •• A ball moves in a straight line (the x -axis). The graph in **Fig. E2.9** shows this ball's velocity as a function of time. (a) What are the ball's average speed and average velocity during the first 3.0 s? (b) Suppose that the ball moved in such a way that the graph segment after 2.0 s was -3.0 m/s instead of $+3.0$ m/s. Find the ball's average speed and average velocity in this case.

Figure E2.9



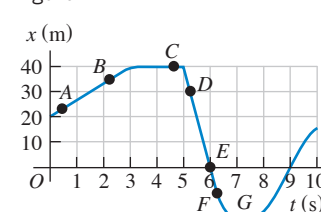
2.10 •• A physics professor leaves her house and walks along the sidewalk toward campus. After 5 min it starts to rain, and she returns home. Her distance from her house as a function of time is shown in **Fig. E2.10**. At which of the labeled points is her velocity (a) zero? (b) constant and positive? (c) constant and negative? (d) increasing in magnitude? (e) decreasing in magnitude?

Figure E2.10



2.11 •• A test car travels in a straight line along the x -axis. The graph in **Fig. E2.11** shows the car's position x as a function of time. Find its instantaneous velocity at points A through G.

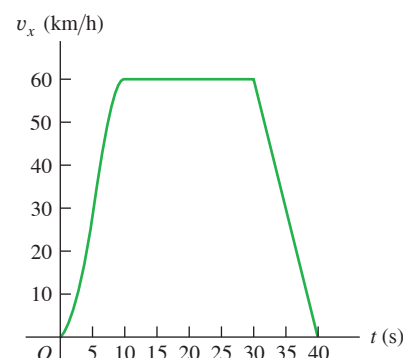
Figure E2.11



Section 2.3 Average and Instantaneous Acceleration

2.12 • **Figure E2.12** shows the velocity of a solar-powered car as a function of time. The driver accelerates from a stop sign, cruises for 20 s at a constant speed of 60 km/h, and then brakes to come to a stop 40 s after leaving the stop sign. (a) Compute the average acceleration during these time intervals: (i) $t = 0$ to $t = 10$ s; (ii) $t = 30$ s to $t = 40$ s; (iii) $t = 10$ s to $t = 30$ s; (iv) $t = 0$ to $t = 40$ s. (b) What is the instantaneous acceleration at $t = 20$ s and at $t = 35$ s?

Figure E2.12



2.13 • CALC A turtle crawls along a straight line, which we'll call the x -axis with the positive direction to the right. The equation for the turtle's position as a function of time is $x(t) = 50.0 \text{ cm} + (2.00 \text{ cm/s})t - (0.0625 \text{ cm/s}^2)t^2$. (a) Find the turtle's initial velocity, initial position, and initial acceleration. (b) At what time t is the velocity of the turtle zero? (c) How long after starting does it take the turtle to return to its starting point? (d) At what times t is the turtle a distance of 10.0 cm from its starting point? What is the velocity (magnitude and direction) of the turtle at each of those times? (e) Sketch graphs of x versus t , v_x versus t , and a_x versus t , for the time interval $t = 0$ to $t = 40 \text{ s}$.

2.14 •• CALC A race car starts from rest and travels east along a straight and level track. For the first 5.0 s of the car's motion, the eastward component of the car's velocity is given by $v_x(t) = (0.860 \text{ m/s}^3)t^2$. What is the acceleration of the car when $v_x = 12.0 \text{ m/s}$?

2.15 • CALC A car's velocity as a function of time is given by $v_x(t) = \alpha + \beta t^2$, where $\alpha = 3.00 \text{ m/s}$ and $\beta = 0.100 \text{ m/s}^3$. (a) Calculate the average acceleration for the time interval $t = 0$ to $t = 5.00 \text{ s}$. (b) Calculate the instantaneous acceleration for $t = 0$ and $t = 5.00 \text{ s}$. (c) Draw v_x - t and a_x - t graphs for the car's motion between $t = 0$ and $t = 5.00 \text{ s}$.

2.16 • An astronaut has left the International Space Station to test a new space scooter. Her partner measures the following velocity changes, each taking place in a 10 s interval. What are the magnitude, the algebraic sign, and the direction of the average acceleration in each interval? Assume that the positive direction is to the right. (a) At the beginning of the interval, the astronaut is moving toward the right along the x -axis at 15.0 m/s, and at the end of the interval she is moving toward the right at 5.0 m/s. (b) At the beginning she is moving toward the left at 5.0 m/s, and at the end she is moving toward the left at 15.0 m/s. (c) At the beginning she is moving toward the right at 15.0 m/s, and at the end she is moving toward the left at 15.0 m/s.

2.17 •• CALC The position of the front bumper of a test car under microprocessor control is given by $x(t) = 2.17 \text{ m} + (4.80 \text{ m/s}^2)t^2 - (0.100 \text{ m/s}^6)t^6$. (a) Find its position and acceleration at the instants when the car has zero velocity. (b) Draw x - t , v_x - t , and a_x - t graphs for the motion of the bumper between $t = 0$ and $t = 2.00 \text{ s}$.

Section 2.4 Motion with Constant Acceleration

2.18 • Estimate the distance in feet that your car travels on the entrance ramp to a freeway as it accelerates from 30 mph to the freeway speed of 70 mph. During this motion what is the average acceleration of the car?

2.19 •• An antelope moving with constant acceleration covers the distance between two points 70.0 m apart in 6.00 s. Its speed as it passes the second point is 15.0 m/s. What are (a) its speed at the first point and (b) its acceleration?

2.20 •• A Tennis Serve. In the fastest measured tennis serve, the ball left the racquet at 73.14 m/s. A served tennis ball is typically in contact with the racquet for 30.0 ms and starts from rest. Assume constant acceleration. (a) What was the ball's acceleration during this serve? (b) How far did the ball travel during the serve?

2.21 • A Fast Pitch. The fastest measured pitched baseball left the pitcher's hand at a speed of 45.0 m/s. If the pitcher was in contact with the ball over a distance of 1.50 m and produced constant acceleration, (a) what acceleration did he give the ball, and (b) how much time did it take him to pitch it?

2.22 • You are traveling on an interstate highway at the posted speed limit of 70 mph when you see that the traffic in front of you has stopped due to an accident up ahead. You step on your brakes to slow down as quickly as possible. (a) Estimate how many seconds it takes you to slow down to 30 mph. What is the magnitude of the average acceleration of the car while it is slowing down? With this same average acceleration,

(b) how much longer would it take you to stop, and (c) what total distance would you travel from when you first apply the brakes until the car stops?

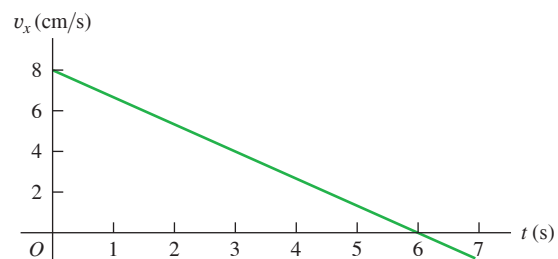
2.23 •• BIO Automobile Airbags. The human body can survive an acceleration trauma incident (sudden stop) if the magnitude of the acceleration is less than 250 m/s^2 . If you are in an automobile accident with an initial speed of 105 km/h (65 mi/h) and are stopped by an airbag that inflates from the dashboard, over what minimum distance must the airbag stop you for you to survive the crash?

2.24 • Entering the Freeway. A car sits on an entrance ramp to a freeway, waiting for a break in the traffic. Then the driver accelerates with constant acceleration along the ramp and onto the freeway. The car starts from rest, moves in a straight line, and has a speed of 20 m/s (45 mi/h) when it reaches the end of the 120-m-long ramp. (a) What is the acceleration of the car? (b) How much time does it take the car to travel the length of the ramp? (c) The traffic on the freeway is moving at a constant speed of 20 m/s. What distance does the traffic travel while the car is moving the length of the ramp?

2.25 • BIO Airbag Injuries. During an auto accident, the vehicle's airbags deploy and slow down the passengers more gently than if they had hit the windshield or steering wheel. According to safety standards, airbags produce a maximum acceleration of $60g$ that lasts for only 36 ms (or less). How far (in meters) does a person travel in coming to a complete stop in 36 ms at a constant acceleration of $60g$?

2.26 •• A cat walks in a straight line, which we shall call the x -axis, with the positive direction to the right. As an observant physicist, you make measurements of this cat's motion and construct a graph of the feline's velocity as a function of time (**Fig. E2.26**). (a) Find the cat's velocity at $t = 4.0 \text{ s}$ and at $t = 7.0 \text{ s}$. (b) What is the cat's acceleration at $t = 3.0 \text{ s}$? At $t = 6.0 \text{ s}$? At $t = 7.0 \text{ s}$? (c) What distance does the cat move during the first 4.5 s? From $t = 0$ to $t = 7.0 \text{ s}$? (d) Assuming that the cat started at the origin, sketch clear graphs of the cat's acceleration and position as functions of time.

Figure E2.26



2.27 • BIO Are We Martians? It has been suggested, and not facetiously, that life might have originated on Mars and been carried to the earth when a meteor hit Mars and blasted pieces of rock (perhaps containing primitive life) free of the Martian surface. Astronomers know that many Martian rocks have come to the earth this way. (For instance, search the Internet for "ALH 84001.") One objection to this idea is that microbes would have had to undergo an enormous lethal acceleration during the impact. Let us investigate how large such an acceleration might be. To escape Mars, rock fragments would have to reach its escape velocity of 5.0 km/s, and that would most likely happen over a distance of about 4.0 m during the meteor impact. (a) What would be the acceleration (in m/s^2 and g 's) of such a rock fragment, if the acceleration is constant? (b) How long would this acceleration last? (c) In tests, scientists have found that over 40% of *Bacillus subtilis* bacteria survived after an acceleration of $450,000g$. In light of your answer to part (a), can we rule out the hypothesis that life might have been blasted from Mars to the earth?

2.28 • Two cars, A and B, move along the x -axis. **Figure E2.28** is a graph of the positions of A and B versus time. (a) In motion diagrams (like Figs. 2.13b and 2.14b), show the position, velocity, and acceleration of each of the two cars at $t = 0$, $t = 1$ s, and $t = 3$ s. (b) At what time(s), if any, do A and B have the same position? (c) Graph velocity versus time for both A and B. (d) At what time(s), if any, do A and B have the same velocity? (e) At what time(s), if any, does car A pass car B? (f) At what time(s), if any, does car B pass car A?

Figure E2.28

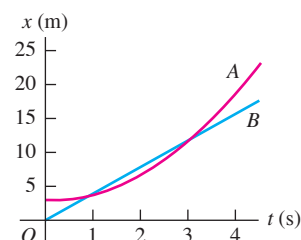
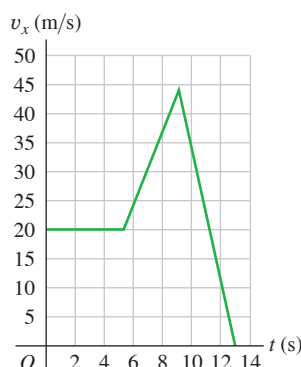


Figure E2.29



2.29 •• The graph in **Fig. E2.29** shows the velocity of a motorcycle police officer plotted as a function of time. (a) Find the instantaneous acceleration at $t = 3$ s, $t = 7$ s, and $t = 11$ s. (b) How far does the officer go in the first 5 s? The first 9 s? The first 13 s?

2.30 •• A small block has constant acceleration as it slides down a frictionless incline. The block is released from rest at the top of the incline, and its speed after it has traveled 6.80 m to the bottom of the incline is 3.80 m/s. What is the speed of the block when it is 3.40 m from the top of the incline?

2.31 •• (a) If a flea can jump straight up to a height of 0.440 m, what is its initial speed as it leaves the ground? (b) How long is it in the air?

2.32 •• A small rock is thrown vertically upward with a speed of 22.0 m/s from the edge of the roof of a 30.0-m-tall building. The rock doesn't hit the building on its way back down and lands on the street below. Ignore air resistance. (a) What is the speed of the rock just before it hits the street? (b) How much time elapses from when the rock is thrown until it hits the street?

Section 2.5 Freely Falling Objects

2.33 • A juggler throws a bowling pin straight up with an initial speed of 8.20 m/s. How much time elapses until the bowling pin returns to the juggler's hand?

2.34 •• You throw a glob of putty straight up toward the ceiling, which is 3.60 m above the point where the putty leaves your hand. The initial speed of the putty as it leaves your hand is 9.50 m/s. (a) What is the speed of the putty just before it strikes the ceiling? (b) How much time from when it leaves your hand does it take the putty to reach the ceiling?

2.35 •• A tennis ball on Mars, where the acceleration due to gravity is $0.379g$ and air resistance is negligible, is hit directly upward and returns to the same level 8.5 s later. (a) How high above its original point did the ball go? (b) How fast was it moving just after it was hit? (c) Sketch graphs of the ball's vertical position, vertical velocity, and vertical acceleration as functions of time while it's in the Martian air.

2.36 •• Estimate the maximum height in feet that you can throw a baseball straight up. (a) For this height, how long after the ball leaves your hand does it return to your hand? (b) Estimate the distance in feet that the ball moves while you are throwing it—that is, the distance from where the ball is when you start your throw until it leaves your hand. Calculate

the average acceleration in m/s^2 that the ball has while it is being thrown, as it moves from rest to the point where it leaves your hand.

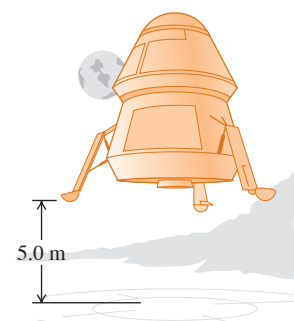
2.37 •• A rock is thrown straight up with an initial speed of 24.0 m/s. Neglect air resistance. (a) At $t = 1.0$ s, what are the directions of the velocity and acceleration of the rock? Is the speed of the rock increasing or decreasing? (b) At $t = 3.0$ s, what are the directions of the velocity and acceleration of the rock? Is the speed of the rock increasing or decreasing?

2.38 •• A brick is dropped (zero initial speed) from the roof of a building. The brick strikes the ground in 1.90 s. You may ignore air resistance, so the brick is in free fall. (a) How tall, in meters, is the building? (b) What is the magnitude of the brick's velocity just before it reaches the ground? (c) Sketch a_y - t , v_y - t , and y - t graphs for the motion of the brick.

2.39 •• A Simple Reaction-Time Test. A meter stick is held vertically above your hand, with the lower end between your thumb and first finger. When you see the meter stick released, you grab it with those two fingers. You can calculate your reaction time from the distance the meter stick falls, read directly from the point where your fingers grabbed it. (a) Derive a relationship for your reaction time in terms of this measured distance, d . (b) If the measured distance is 17.6 cm, what is your reaction time?

2.40 •• Touchdown on the Moon. **Figure E2.40**

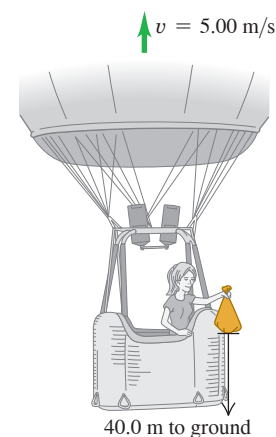
A lunar lander is making its descent to Moon Base I (**Fig. E2.40**). The lander descends slowly under the retro-thrust of its descent engine. The engine is cut off when the lander is 5.0 m above the surface and has a downward speed of 0.8 m/s. With the engine off, the lander is in free fall. What is the speed of the lander just before it touches the surface? The acceleration due to gravity on the moon is 1.6 m/s^2 .



2.41 •• Launch Failure. A 7500 kg rocket blasts off vertically from the launch pad with a constant upward acceleration of 2.25 m/s^2 and feels no appreciable air resistance. When it has reached a height of 525 m, its engines suddenly fail; the only force acting on it is now gravity. (a) What is the maximum height this rocket will reach above the launch pad? (b) How much time will elapse after engine failure before the rocket comes crashing down to the launch pad, and how fast will it be moving just before it crashes? (c) Sketch a_y - t , v_y - t , and y - t graphs of the rocket's motion from the instant of blast-off to the instant just before it strikes the launch pad.

Figure E2.42

2.42 •• A hot-air balloonist, rising vertically with a constant velocity of magnitude 5.00 m/s, releases a sandbag at an instant when the balloon is 40.0 m above the ground (**Fig. E2.42**). After the sandbag is released, it is in free fall. (a) Compute the position and velocity of the sandbag at 0.250 s and 1.00 s after its release. (b) How many seconds after its release does the bag strike the ground? (c) With what magnitude of velocity does it strike the ground? (d) What is the greatest height above the ground that the sandbag reaches? (e) Sketch a_y - t , v_y - t , and y - t graphs for the motion.



2.43 • You throw a rock straight up and find that it returns to your hand 3.60 s after it left your hand. Neglect air resistance. What was the maximum height above your hand that the rock reached?

2.44 • An egg is thrown nearly vertically upward from a point near the cornice of a tall building. The egg just misses the cornice on the way down and passes a point 30.0 m below its starting point 5.00 s after it leaves the thrower's hand. Ignore air resistance. (a) What is the initial speed of the egg? (b) How high does it rise above its starting point? (c) What is the magnitude of its velocity at the highest point? (d) What are the magnitude and direction of its acceleration at the highest point? (e) Sketch a_y - t , v_y - t , and y - t graphs for the motion of the egg.

2.45 •• A 15 kg rock is dropped from rest on the earth and reaches the ground in 1.75 s. When it is dropped from the same height on Saturn's satellite Enceladus, the rock reaches the ground in 18.6 s. What is the acceleration due to gravity on Enceladus?

2.46 • A large boulder is ejected vertically upward from a volcano with an initial speed of 40.0 m/s. Ignore air resistance. (a) At what time after being ejected is the boulder moving at 20.0 m/s upward? (b) At what time is it moving at 20.0 m/s downward? (c) When is the displacement of the boulder from its initial position zero? (d) When is the velocity of the boulder zero? (e) What are the magnitude and direction of the acceleration while the boulder is (i) moving upward? (ii) Moving downward? (iii) At the highest point? (f) Sketch a_y - t , v_y - t , and y - t graphs for the motion.

2.47 •• You throw a small rock straight up from the edge of a highway bridge that crosses a river. The rock passes you on its way down, 6.00 s after it was thrown. What is the speed of the rock just before it reaches the water 28.0 m below the point where the rock left your hand? Ignore air resistance.

Section 2.6 Velocity and Position by Integration

2.48 •• Consider the motion described by the v_x - t graph of Fig. E2.26. (a) Calculate the area under the graph between $t = 0$ and $t = 6.0$ s. (b) For the time interval $t = 0$ to $t = 6.0$ s, what is the magnitude of the average velocity of the cat? (c) Use constant-acceleration equations to calculate the distance the cat travels in this time interval. How does your result compare to the area you calculated in part (a)?

2.49 • CALC A rocket starts from rest and moves upward from the surface of the earth. For the first 10.0 s of its motion, the vertical acceleration of the rocket is given by $a_y = (2.80 \text{ m/s}^3)t$, where the $+y$ -direction is upward. (a) What is the height of the rocket above the surface of the earth at $t = 10.0$ s? (b) What is the speed of the rocket when it is 325 m above the surface of the earth?

2.50 •• CALC A small object moves along the x -axis with acceleration $a_x(t) = -(0.0320 \text{ m/s}^3)(15.0 \text{ s} - t)$. At $t = 0$ the object is at $x = -14.0$ m and has velocity $v_{0x} = 8.00$ m/s. What is the x -coordinate of the object when $t = 10.0$ s?

2.51 •• CALC The acceleration of a motorcycle is given by $a_x(t) = At - Bt^2$, where $A = 1.50 \text{ m/s}^3$ and $B = 0.120 \text{ m/s}^4$. The motorcycle is at rest at the origin at time $t = 0$. (a) Find its position and velocity as functions of time. (b) Calculate the maximum velocity it attains.

2.52 •• CALC The acceleration of a bus is given by $a_x(t) = \alpha t$, where $\alpha = 1.2 \text{ m/s}^3$. (a) If the bus's velocity at time $t = 1.0$ s is 5.0 m/s, what is its velocity at time $t = 2.0$ s? (b) If the bus's position at time $t = 1.0$ s is 6.0 m, what is its position at time $t = 2.0$ s? (c) Sketch a_y - t , v_y - t , and x - t graphs for the motion.

PROBLEMS

2.53 • BIO A typical male sprinter can maintain his maximum acceleration for 2.0 s, and his maximum speed is 10 m/s. After he reaches this maximum speed, his acceleration becomes zero, and then he runs at constant speed. Assume that his acceleration is constant during the first

2.0 s of the race, that he starts from rest, and that he runs in a straight line. (a) How far has the sprinter run when he reaches his maximum speed? (b) What is the magnitude of his average velocity for a race of these lengths: (i) 50.0 m; (ii) 100.0 m; (iii) 200.0 m?

2.54 • CALC A lunar lander is descending toward the moon's surface. Until the lander reaches the surface, its height above the surface of the moon is given by $y(t) = b - ct + dt^2$, where $b = 800$ m is the initial height of the lander above the surface, $c = 60.0$ m/s, and $d = 1.05 \text{ m/s}^2$. (a) What is the initial velocity of the lander, at $t = 0$? (b) What is the velocity of the lander just before it reaches the lunar surface?

2.55 •• Earthquake Analysis. Earthquakes produce several types of shock waves. The most well known are the P-waves (P for *primary* or *pressure*) and the S-waves (S for *secondary* or *shear*). In the earth's crust, P-waves travel at about 6.5 km/s and S-waves move at about 3.5 km/s. The time delay between the arrival of these two waves at a seismic recording station tells geologists how far away an earthquake occurred. If the time delay is 33 s, how far from the seismic station did the earthquake occur?

2.56 •• You throw a small rock straight up with initial speed V_0 from the edge of the roof of a building that is a distance H above the ground. The rock travels upward to a maximum height in time T_{max} , misses the edge of the roof on its way down, and reaches the ground in time T_{total} after it was thrown. Neglect air resistance. If the total time the rock is in the air is three times the time it takes it to reach its maximum height, so $T_{\text{total}} = 3T_{\text{max}}$, then in terms of H what must be the value of V_0 ?

2.57 •• A rocket carrying a satellite is accelerating straight up from the earth's surface. At 1.15 s after liftoff, the rocket clears the top of its launch platform, 63 m above the ground. After an additional 4.75 s, it is 1.00 km above the ground. Calculate the magnitude of the average velocity of the rocket for (a) the 4.75 s part of its flight and (b) the first 5.90 s of its flight.

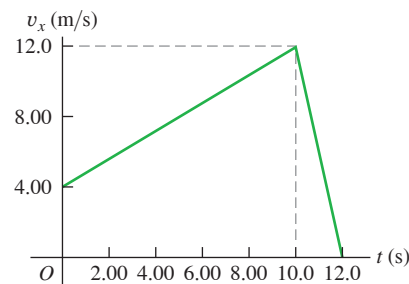
2.58 •• A block moving on a horizontal surface is at $x = 0$ when $t = 0$ and is sliding east with a speed of 12.0 m/s. Because of a net force acting on the block, it has a constant acceleration with direction west and magnitude 2.00 m/s^2 . The block travels east, slows down, reverses direction, and then travels west with increasing speed. (a) At what value of t is the block again at $x = 0$? (b) What is the maximum distance east of $x = 0$ that the block reaches, and how long does it take the block to reach this point?

2.59 •• A block is sliding with constant acceleration down an incline. The block starts from rest at $t = 0$ and has speed 3.00 m/s after it has traveled a distance 8.00 m from its starting point. (a) What is the speed of the block when it is a distance of 16.0 m from its $t = 0$ starting point? (b) How long does it take the block to slide 16.0 m from its starting point?

2.60 •• A subway train starts from rest at a station and accelerates at a rate of 1.60 m/s^2 for 14.0 s. It runs at constant speed for 70.0 s and slows down at a rate of 3.50 m/s^2 until it stops at the next station. Find the total distance covered.

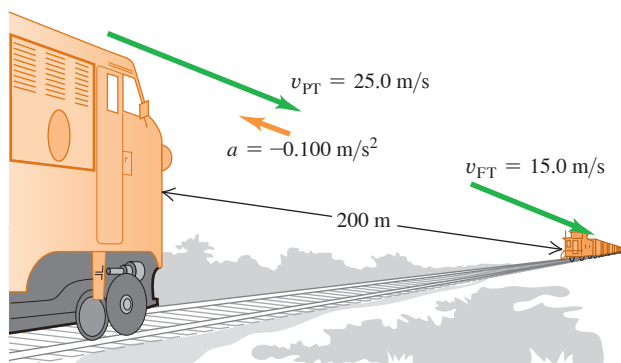
2.61 • A gazelle is running in a straight line (the x -axis). The graph in Fig. P2.61 shows this animal's velocity as a function of time. During the first 12.0 s, find (a) the total distance moved and (b) the displacement of the gazelle. (c) Sketch an a_x - t graph showing this gazelle's acceleration as a function of time for the first 12.0 s.

Figure P2.61



2.62 •• Collision. The engineer of a passenger train traveling at 25.0 m/s sights a freight train whose caboose is 200 m ahead on the same track (**Fig. P2.62**). The freight train is traveling at 15.0 m/s in the same direction as the passenger train. The engineer of the passenger train immediately applies the brakes, causing a constant acceleration of 0.100 m/s^2 in a direction opposite to the train's velocity, while the freight train continues with constant speed. Take $x = 0$ at the location of the front of the passenger train when the engineer applies the brakes. (a) Will the cows nearby witness a collision? (b) If so, where will it take place? (c) On a single graph, sketch the positions of the front of the passenger train and the back of the freight train.

Figure P2.62



2.63 ••• A ball starts from rest and rolls down a hill with uniform acceleration, traveling 200 m during the second 5.0 s of its motion. How far did it roll during the first 5.0 s of motion?

2.64 •• A rock moving in the $+x$ -direction with speed 16.0 m/s has a net force applied to it at time $t = 0$, and this produces a constant acceleration in the $-x$ -direction that has magnitude 4.00 m/s^2 . For what three times t after the force is applied is the rock a distance of 24.0 m from its position at $t = 0$? For each of these three values of t , what is the velocity (magnitude and direction) of the rock?

2.65 • A car and a truck start from rest at the same instant, with the car initially at some distance behind the truck. The truck has a constant acceleration of 2.10 m/s^2 , and the car has an acceleration of 3.40 m/s^2 . The car overtakes the truck after the truck has moved 60.0 m. (a) How much time does it take the car to overtake the truck? (b) How far was the car behind the truck initially? (c) What is the speed of each when they are abreast? (d) On a single graph, sketch the position of each vehicle as a function of time. Take $x = 0$ at the initial location of the truck.

2.66 •• You are standing at rest at a bus stop. A bus moving at a constant speed of 5.00 m/s passes you. When the rear of the bus is 12.0 m past you, you realize that it is your bus, so you start to run toward it with a constant acceleration of 0.960 m/s^2 . How far would you have to run before you catch up with the rear of the bus, and how fast must you be running then? Would an average college student be physically able to accomplish this?

2.67 •• A sprinter runs a 100 m dash in 12.0 s. She starts from rest with a constant acceleration a_x for 3.0 s and then runs with constant speed for the remainder of the race. What is the value of a_x ?

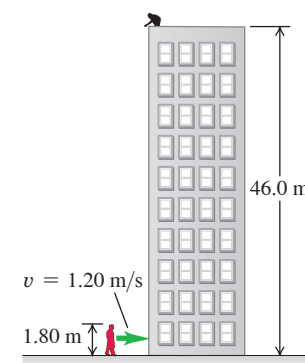
2.68 •• CALC An object's velocity is measured to be $v_x(t) = \alpha - \beta t^2$, where $\alpha = 4.00 \text{ m/s}$ and $\beta = 2.00 \text{ m/s}^3$. At $t = 0$ the object is at $x = 0$. (a) Calculate the object's position and acceleration as functions of time. (b) What is the object's maximum positive displacement from the origin?

2.69 •• CALC An object is moving along the x -axis. At $t = 0$ it is at $x = 0$. Its x -component of velocity v_x as a function of time is given by $v_x(t) = \alpha t - \beta t^3$, where $\alpha = 8.0 \text{ m/s}^2$ and $\beta = 4.0 \text{ m/s}^4$. (a) At what

nonzero time t is the object again at $x = 0$? (b) At the time calculated in part (a), what are the velocity and acceleration of the object (magnitude and direction)?

2.70 • Egg Drop. You are on the roof of the physics building, 46.0 m above the ground (**Fig. P2.70**). Your physics professor, who is 1.80 m tall, is walking alongside the building at a constant speed of 1.20 m/s. If you wish to drop an egg on your professor's head, where should the professor be when you release the egg? Assume that the egg is in free fall.

Figure P2.70



2.71 ••• CALC The acceleration of a particle is given by $a_x(t) = -2.00 \text{ m/s}^2 + (3.00 \text{ m/s}^3)t$. (a) Find the initial velocity v_{0x} such that the particle will have the same x -coordinate at $t = 4.00 \text{ s}$ as it had at $t = 0$. (b) What will be the velocity at $t = 4.00 \text{ s}$?

2.72 •• A small rock is thrown straight up with initial speed v_0 from the edge of the roof of a building with height H . The rock travels upward and then downward to the ground at the base of the building. Let $+y$ be upward, and neglect air resistance. (a) For the rock's motion from the roof to the ground, what is the vertical component v_{av-y} of its average velocity? Is this quantity positive or negative? Explain. (b) What does your expression for v_{av-y} give in the limit that H is zero? Explain. (c) Show that your result in part (a) agrees with Eq. (2.10).

2.73 •• A watermelon is dropped from the edge of the roof of a building and falls to the ground. You are standing on the sidewalk and see the watermelon falling when it is 30.0 m above the ground. Then 1.50 s after you first spot it, the watermelon lands at your feet. What is the height of the building? Neglect air resistance.

2.74 ••• A flowerpot falls off a windowsill and passes the window of the story below. Ignore air resistance. It takes the pot 0.380 s to pass from the top to the bottom of this window, which is 1.90 m high. How far is the top of the window below the windowsill from which the flowerpot fell?

2.75 ••• Look Out Below. Sam heaves a 16 lb shot straight up, giving it a constant upward acceleration from rest of 35.0 m/s^2 for 64.0 cm. He releases it 2.20 m above the ground. Ignore air resistance. (a) What is the speed of the shot when Sam releases it? (b) How high above the ground does it go? (c) How much time does he have to get out of its way before it returns to the height of the top of his head, 1.83 m above the ground?

2.76 ••• A Multistage Rocket. In the first stage of a two-stage rocket, the rocket is fired from the launch pad starting from rest but with a constant acceleration of 3.50 m/s^2 upward. At 25.0 s after launch, the second stage fires for 10.0 s, which boosts the rocket's velocity to 132.5 m/s upward at 35.0 s after launch. This firing uses up all of the fuel, however, so after the second stage has finished firing, the only force acting on the rocket is gravity. Ignore air resistance. (a) Find the maximum height that the stage-two rocket reaches above the launch pad. (b) How much time after the end of the stage-two firing will it take for the rocket to fall back to the launch pad? (c) How fast will the stage-two rocket be moving just as it reaches the launch pad?

2.77 •• Two stones are thrown vertically upward from the ground, one with three times the initial speed of the other. (a) If the faster stone takes 10 s to return to the ground, how long will it take the slower stone to return? (b) If the slower stone reaches a maximum height of H , how high (in terms of H) will the faster stone go? Assume free fall.

2.78 •• During your summer internship for an aerospace company, you are asked to design a small research rocket. The rocket is to be launched from rest from the earth's surface and is to reach a maximum height of 960 m above the earth's surface. The rocket's engines give the rocket an upward acceleration of 16.0 m/s^2 during the time T that they fire. After the engines shut off, the rocket is in free fall. Ignore air resistance. What must be the value of T in order for the rocket to reach the required altitude?

2.79 •• A helicopter carrying Dr. Evil takes off with a constant upward acceleration of 5.0 m/s^2 . Secret agent Austin Powers jumps on just as the helicopter lifts off the ground. After the two men struggle for 10.0 s, Powers shuts off the engine and steps out of the helicopter. Assume that the helicopter is in free fall after its engine is shut off, and ignore the effects of air resistance. (a) What is the maximum height above ground reached by the helicopter? (b) Powers deploys a jet pack strapped on his back 7.0 s after leaving the helicopter, and then he has a constant downward acceleration with magnitude 2.0 m/s^2 . How far is Powers above the ground when the helicopter crashes into the ground?

2.80 •• Cliff Height. You are climbing in the High Sierra when you suddenly find yourself at the edge of a fog-shrouded cliff. To find the height of this cliff, you drop a rock from the top; 8.00 s later you hear the sound of the rock hitting the ground at the foot of the cliff. (a) If you ignore air resistance, how high is the cliff if the speed of sound is 330 m/s ? (b) Suppose you had ignored the time it takes the sound to reach you. In that case, would you have overestimated or underestimated the height of the cliff? Explain.

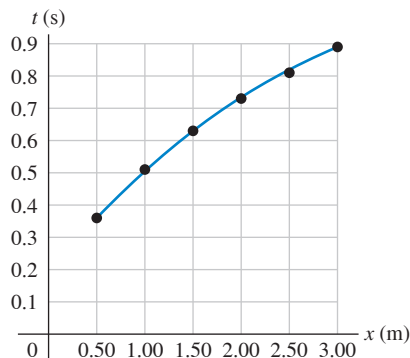
2.81 •• CALC An object is moving along the x -axis. At $t = 0$ it has velocity $v_{0x} = 20.0 \text{ m/s}$. Starting at time $t = 0$ it has acceleration $a_x = -Ct$, where C has units of m/s^3 . (a) What is the value of C if the object stops in 8.00 s after $t = 0$? (b) For the value of C calculated in part (a), how far does the object travel during the 8.00 s?

2.82 •• A ball is thrown straight up from the ground with speed v_0 . At the same instant, a second ball is dropped from rest from a height H , directly above the point where the first ball was thrown upward. There is no air resistance. (a) Find the time at which the two balls collide. (b) Find the value of H in terms of v_0 and g such that at the instant when the balls collide, the first ball is at the highest point of its motion.

2.83 • CALC Cars A and B travel in a straight line. The distance of A from the starting point is given as a function of time by $x_A(t) = \alpha t + \beta t^2$, with $\alpha = 2.60 \text{ m/s}$ and $\beta = 1.20 \text{ m/s}^2$. The distance of B from the starting point is $x_B(t) = \gamma t^2 - \delta t^3$, with $\gamma = 2.80 \text{ m/s}^2$ and $\delta = 0.20 \text{ m/s}^3$. (a) Which car is ahead just after the two cars leave the starting point? (b) At what time(s) are the cars at the same point? (c) At what time(s) is the distance from A to B neither increasing nor decreasing? (d) At what time(s) do A and B have the same acceleration?

2.84 •• DATA In your physics lab you release a small glider from rest at various points on a long, frictionless air track that is inclined at an angle θ above the horizontal. With an electronic photocell, you measure the time t it takes the glider to slide a distance x from the release point to the bottom of the track. Your measurements are given in **Fig. P2.84**, which shows a second-order polynomial (quadratic) fit to the plotted data. You are asked to find the glider's acceleration, which is assumed to be constant. There is some error in each measurement, so instead of using a single set of x and t values, you can be more accurate if you use graphical methods and obtain your measured value of the acceleration from the graph. (a) How can you re-graph the data so that the data points fall close to a straight line? (*Hint:* You might want to plot x or t , or both, raised to some power.) (b) Construct the graph you described in part (a) and find the equation for the straight line that is the best fit to the data points. (c) Use the straight-line fit from part (b) to calculate the acceleration of the glider. (d) The glider is released at a distance $x = 1.35 \text{ m}$ from the bottom of the track. Use the acceleration value you obtained in part (c) to calculate the speed of the glider when it reaches the bottom of the track.

Figure P2.84



2.85 •• DATA In a physics lab experiment, you release a small steel ball at various heights above the ground and measure the ball's speed just before it strikes the ground. You plot your data on a graph that has the release height (in meters) on the vertical axis and the square of the final speed (in m^2/s^2) on the horizontal axis. In this graph your data points lie close to a straight line. (a) Using $g = 9.80 \text{ m/s}^2$ and ignoring the effect of air resistance, what is the numerical value of the slope of this straight line? (Include the correct units.) The presence of air resistance reduces the magnitude of the downward acceleration, and the effect of air resistance increases as the speed of the object increases. You repeat the experiment, but this time with a tennis ball as the object being dropped. Air resistance now has a noticeable effect on the data. (b) Is the final speed for a given release height higher than, lower than, or the same as when you ignored air resistance? (c) Is the graph of the release height versus the square of the final speed still a straight line? Sketch the qualitative shape of the graph when air resistance is present.

2.86 ••• DATA A model car starts from rest and travels in a straight line. A smartphone mounted on the car has an app that transmits the magnitude of the car's acceleration (measured by an accelerometer) every second. The results are given in the table:

Time (s)	Acceleration (m/s^2)
0	5.95
1.00	5.52
2.00	5.08
3.00	4.55
4.00	3.96
5.00	3.40

Each measured value has some experimental error. (a) Plot acceleration versus time and find the equation for the straight line that gives the best fit to the data. (b) Use the equation for $a(t)$ that you found in part (a) to calculate $v(t)$, the speed of the car as a function of time. Sketch the graph of v versus t . Is this graph a straight line? (c) Use your result from part (b) to calculate the speed of the car at $t = 5.00 \text{ s}$. (d) Calculate the distance the car travels between $t = 0$ and $t = 5.00 \text{ s}$.

CHALLENGE PROBLEMS

2.87 ••• In the vertical jump, an athlete starts from a crouch and jumps upward as high as possible. Even the best athletes spend little more than 1.00 s in the air (their "hang time"). Treat the athlete as a particle and let y_{max} be his maximum height above the floor. To explain why he seems to hang in the air, calculate the ratio of the time he is above $y_{\text{max}}/2$ to the time it takes him to go from the floor to that height. Ignore air resistance.

2.88 ••• Catching the Bus. A student is running at her top speed of 5.0 m/s to catch a bus, which is stopped at the bus stop. When the student is still 40.0 m from the bus, it starts to pull away, moving with a constant acceleration of 0.170 m/s^2 . (a) For how much time and what distance does the student have to run at 5.0 m/s before she overtakes the bus? (b) When she reaches the bus, how fast is the bus traveling? (c) Sketch an x - t graph for both the student and the bus. Take $x = 0$ at the initial position of the student. (d) The equations you used in part (a) to find the time have a second solution, corresponding to a later time for which the student and bus are again at the same place if they continue their specified motions. Explain the significance of this second solution. How fast is the bus traveling at this point? (e) If the student's top speed is 3.5 m/s, will she catch the bus? (f) What is the *minimum* speed the student must have to just catch up with the bus? For what time and what distance does she have to run in that case?

2.89 ••• A ball is thrown straight up from the edge of the roof of a building. A second ball is dropped from the roof 1.00 s later. Ignore air resistance. (a) If the height of the building is 20.0 m, what must the initial speed of the first ball be if both are to hit the ground at the same time? On the same graph, sketch the positions of both balls as a function of time, measured from when the first ball is thrown. Consider the same situation, but now let the initial speed v_0 of the first ball be given and treat the height h of the building as an unknown. (b) What must the height of the building be for both balls to reach the ground at the same time if (i) v_0 is 6.0 m/s and (ii) v_0 is 9.5 m/s? (c) If v_0 is greater than some value v_{max} , no value of h exists that allows both balls to hit the ground at the same time. Solve for v_{max} . The value v_{max} has a simple physical interpretation. What is it? (d) If v_0 is less than some value v_{min} , no value of h exists that allows both balls to hit the ground at the same time. Solve for v_{min} . The value v_{min} also has a simple physical interpretation. What is it?

MCAT-STYLE PASSAGE PROBLEMS

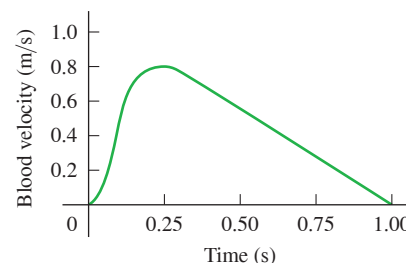
BIO Blood Flow in the Heart. The human circulatory system is closed—that is, the blood pumped out of the left ventricle of the heart into the arteries is constrained to a series of continuous, branching vessels as it passes through the capillaries and then into the veins as it returns to the heart. The blood in each of the heart's four chambers comes briefly to rest before it is ejected by contraction of the heart muscle.

2.90 If the contraction of the left ventricle lasts 250 ms and the speed of blood flow in the aorta (the large artery leaving the heart) is 0.80 m/s at the end of the contraction, what is the average acceleration of a red blood cell as it leaves the heart? (a) 310 m/s^2 ; (b) 31 m/s^2 ; (c) 3.2 m/s^2 ; (d) 0.32 m/s^2 .

2.91 If the aorta (diameter d_a) branches into two equal-sized arteries with a combined area equal to that of the aorta, what is the diameter of one of the branches? (a) $\sqrt{d_a}$; (b) $d_a/\sqrt{2}$; (c) $2d_a$; (d) $d_a/2$.

2.92 The velocity of blood in the aorta can be measured directly with ultrasound techniques. A typical graph of blood velocity versus time during a single heartbeat is shown in **Fig. P2.92**. Which statement is the best interpretation of this graph? (a) The blood flow changes direction at about 0.25 s; (b) the speed of the blood flow begins to decrease at about 0.10 s; (c) the acceleration of the blood is greatest in magnitude at about 0.25 s; (d) the acceleration of the blood is greatest in magnitude at about 0.10 s.

Figure P2.92



ANSWERS

Chapter Opening Question ?

(iii) Acceleration refers to *any* change in velocity, including both speeding up and slowing down.

Key Example VARIATION Problems

VP2.5.1 (a) 4.5 s (b) 81 m

VP2.5.2 (a) 4.80 m/s^2 (b) 375 m

VP2.5.3 (a) 11.1 s (b) 222 m (c) 10.0 m/s

VP2.5.4 (a) $t = 2v_C/(a_T + a_C)$ (b) $x = 2a_T v_C^2/(a_T + a_C)^2$

VP2.7.1 (a) $y = 3.16 \text{ m}$, $v_y = 9.06 \text{ m/s}$, above your hand and moving upward (b) $y = -1.92 \text{ m}$, $v_y = -13.5 \text{ m/s}$, below your hand and moving downward

VP2.7.2 (a) $y = -23.0 \text{ m}$, $v_y = -22.7 \text{ m/s}$ (b) $v_y = -14.9 \text{ m/s}$

VP2.7.3 (a) 8.87 m/s (b) 0.854 s

VP2.7.4 (a) $\sqrt{2gH}$ (b) $\sqrt{gH} = 1/\sqrt{2} = 0.707$ times the speed at which it left your hand (c) $3H/4$

VP2.8.1 (a) 0.398 s and 2.05 s (b) 2.75 s

VP2.8.2 0.447 s

VP2.8.3 (a) two; $t = 0.338 \text{ s}$ (moving upward) and $t = 0.784 \text{ s}$ (moving downward) (b) none

VP2.8.4 (a) $t = v_0/g$ (neither upward nor downward) (b) $t = v_0/2g$ (upward), $t = 3v_0/2g$ (downward)

Bridging Problem

$h = 57.1 \text{ m}$